

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
41	02	INJECTOR INJECTOR SUPPLY	- B BELOW LOWER LIMIT	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.				
41	04	INJECTOR INJECTOR SUPPLY	- B NO SIGNAL	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.				
41	08	INJECTOR INJECTOR SUPPLY	- B SIGNAL NOT PLAUSIBLE	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.				
42	01	INJECTOR INJECTOR I	- EXCEEDED UPPER LIMIT	Engine not working properly, possible power reduction.	Short-circuit to positive.	Check wiring and connections. Replace injector if required.				Only three cylinders running.

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
42	01	INJECTOR INJECTOR I	- EXCEEDED UPPER LIMIT				1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A47 Measure point 2: Injector Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: Injector Pin: 1 Measure point 2: Injector Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A16 Measure point 2: Injector Pin: 1	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Min. value:0.5 Ohm; Max. value: 0.9 Ohm; Typical Value: 0.7 Ohm; 3- Typical Value: 0,1 Ohm;	
42	04	INJECTOR INJECTOR I	- NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring short-circuit.	Check wiring and connections. Replace injector if required.				Only two cylinders running.
42	04	INJECTOR INJECTOR I	- NO SIGNAL				1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A47 Measure point 2: Injector Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: Injector Pin: 1 Measure point 2: Injector Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A16 Measure point 2: Injector Pin: 1	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Min. value:0.5 Ohm; Max. value: 0.9 Ohm; Typical Value: 0.7 Ohm; 3- Typical Value: 0,1 Ohm;	

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
42	08	INJECTOR INJECTOR 1	- SIGNAL NOT PLAUSIBLE	Engine not working properly, possible power reduction.	Injector not working properly.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
42	08	INJECTOR INJECTOR 1	- SIGNAL NOT PLAUSIBLE				1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A47 Measure point 2: Injector Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: Injector Pin: 1 Measure point 2: Injector Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A16 Measure point 2: Injector Pin: 1	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0.1 Ohm; 2- Min. value: 0.5 Ohm; Max. value: 0.9 Ohm; Typical Value: 0.7 Ohm; 3- Typical Value: 0.1 Ohm;	
43	04	INJECTOR INJECTOR 1	- NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring open circuit.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
44	01	INJECTOR INJECTOR 2	- EXCEEDED UPPER LIMIT	Engine not working properly, possible power reduction.	Short-circuit to positive.	Check wiring and connections. Replace injector if required.				Only three cylinders running.

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
44	01	INJECTOR INJECTOR 2	- EXCEEDED UPPER LIMIT				1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A17 Measure point 2: Injector Pin: 1 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A13 Measure point 2: Injector Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: Injector Pin: 1 Measure point 2: Injector Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	
44	04	INJECTOR INJECTOR 2	- NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring short-circuit.	Check wiring and connections. Replace injector if required.				Only two cylinders running.
44	04	INJECTOR INJECTOR 2	- NO SIGNAL				1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A17 Measure point 2: Injector Pin: 1 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A13 Measure point 2: Injector Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: Injector Pin: 1 Measure point 2: Injector Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
44	08	INJECTOR INJECTOR 2	- SIGNAL NOT PLAUSIBLE				1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A17 Measure point 2: Injector Pin: 1 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A13 Measure point 2: Injector Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: Injector Pin: 1 Measure point 2: Injector Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	
44	08	INJECTOR INJECTOR 2	- SIGNAL NOT PLAUSIBLE	Engine not working properly, possible power reduction.	Injector not working properly.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
45	04	INJECTOR INJECTOR 2	- NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring open circuit.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
46	01	INJECTOR INJECTOR 3	- EXCEEDED UPPER LIMIT	Engine not working properly, possible power reduction.	Short-circuit to positive.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
46	04	INJECTOR INJECTOR 3	- NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring short-circuit.	Check wiring and connections. Replace injector if required.				Only two cylinders running.

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
46	08	INJECTOR INJECTOR 3	- SIGNAL NOT PLAUSIBLE	Engine not working properly, possible power reduction.	Injector not working properly.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
47	04	INJECTOR INJECTOR 3	- NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring open circuit.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
47	04	INJECTOR INJECTOR 3	- NO SIGNAL				1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A31 Measure point 2: Injector Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A1 Measure point 2: Injector Pin: 1 3- Measure type: Resistance (Ohm) Measure point 1: Injector Pin: 1 Measure point 2: Injector Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	
48	01	INJECTOR INJECTOR 4	- EXCEEDED UPPER LIMIT	Engine not working properly, possible power reduction.	Short-circuit to positive.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
48	04	INJECTOR INJECTOR 4	- NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring short-circuit.	Check wiring and connections. Replace injector if required.				Only two cylinders running.

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
48	08	INJECTOR INJECTOR 4	- SIGNAL NOT PLAUSIBLE	Engine not working properly, possible power reduction.	Injector not working properly.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
49	04	INJECTOR INJECTOR 4	- NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring open circuit.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
49	04	INJECTOR INJECTOR 4	- NO SIGNAL				1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A46 Measure point 2: Injector Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A1 Measure point 2: Injector Pin: 1 3- Measure type: Resistance (Ohm) Measure point 1: Injector Pin: 1 Measure point 2: Injector Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	
4E	08	VEHICLE CRUISE CONTROL SWITCH UNIT	- SIGNAL NOT PLAUSIBLE	Cruise control / PTO not working.	Press SET+ / SET- and RESUME/ OFF at the same time.	Check correct operation of the switch by reading state parameters.				Replace wiring and connections if state does not change when Cruise Control buttons are pressed.

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
50	01	ELECTRONIC CONTROL UNIT - MAIN RELAY DEFECT	EXCEEDED UPPER LIMIT	Engine does not start, ECU not powered or ECU always powered and EDC off also at key-on.	Main relay interrupted or short-circuit.	Check wiring and connections. Replace relay if required.				
50	02	ELECTRONIC CONTROL UNIT - MAIN RELAY DEFECT	BELOW LOWER LIMIT	Engine does not start, ECU not powered or ECU always powered and EDC off also at key-on.	Main relay interrupted or short-circuit.	Check wiring and connections. Replace relay if required.				
51	01	VEHICLE - MULTIPOSITION SELECTOR / PTO	EXCEEDED UPPER LIMIT	Incorrect PTO operation.	Voltage exceeding max. threshold, short-circuit to positive.	Check wiring and connections. Replace sensor if required.				
51	02	VEHICLE - MULTIPOSITION SELECTOR / PTO	BELOW LOWER LIMIT	Incorrect PTO operation.	Voltage under min. threshold, short-circuit to ground.	Check wiring and connections. Replace sensor if required.				
51	08	VEHICLE - MULTIPOSITION SELECTOR / PTO	SIGNAL NOT PLAUSIBLE	Incorrect PTO operation.	Faulty device.	Check wiring and connections. Replace sensor if required.				
52	04	FUEL PRESSURE - PRESSURE MPROP REGULATOR ERROR	NO SIGNAL	Engine off.	Faulty MPROP.	Check wiring and connections.	Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A49 Measure point 2: ECU Pin: A19	Connector Not connected; Key +15 OFF;	Min. value: 3.2 Ohm; Max. value: 3.6 Ohm; Typical Value: 3.4 Ohm;	High noise.

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
52	08	FUEL PRESSURE - PRESSURE MPROP REGULATOR ERROR	SIGNAL NOT PLAUSIBLE			Check wiring and connections. Replace ECU if required.	Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A49 Measure point 2: ECU Pin: A19	Connector Not connected; Key +15 OFF;	Min. value: 3.2 Ohm; Max. value: 3.6 Ohm; Typical Value: 3.4 Ohm;	
53	01	FUEL PRESSURE - PRESSURE MPROP REGULATOR ERROR (SHORT CIRCUIT TO POSITIVE)	EXCEEDED UPPER LIMIT		Short-circuit to battery, faulty MPROP.	Check wiring and connections. Replace MPROP if required.				
54	01	FUEL PRESSURE - PRESSURE MPROP REGULATOR ERROR (SHORT CIRCUIT TO NEGATIVE)	EXCEEDED UPPER LIMIT		Short-circuit to ground. Faulty MPROP.	Check wiring and connections. Replace MPROP if required.				
56	08	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
5A	01	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	EXCEEDED UPPER LIMIT		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
5A	02	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	BELOW LOWER LIMIT		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
5B	01	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	EXCEEDED UPPER LIMIT		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
5E	01	ENGINE I - FUEL PUMP RELAY	EXCEEDED UPPER LIMIT	Fuel pump on always when engine is off.	Faulty relay, short-circuit to positive in wiring.	Turn key-on: pump must run for approximately 10 seconds (it should hum). Check pump relay if pump remains on. Check wiring if all checks are OK.				

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
5E	02	ENGINE I - FUEL PUMP RELAY	BELOW LOWER LIMIT	Fuel pump not working.	Faulty relay, short-circuit to ground in wiring.	Turn key-on: pump must run for approximately 10 seconds (it should hum). Check the pump relay, protection fuse and wiring if this does not occur.				
5E	04	ENGINE I - FUEL PUMP RELAY	NO SIGNAL	Fuel pump not working.	Faulty relay, wiring interrupted.	Check wiring and connections. Replace relay if required.				
5E	08	ENGINE I - FUEL PUMP RELAY	SIGNAL NOT PLAUSIBLE	Fuel pump not working.	Faulty relay, wiring interrupted.	Check wiring and connections. Replace relay if required.				
5F	01	FUEL PRESSURE - RAIL PRESSURE SENSOR OR SIGNAL ERROR	EXCEEDED UPPER LIMIT		Short-circuit to positive. Faulty sensor. Rail pressure not regular.	Check wiring and connections. Replace sensor if required.				Check DTC I03 error.
5F	02	FUEL PRESSURE - RAIL PRESSURE SENSOR OR SIGNAL ERROR	BELOW LOWER LIMIT		Short-circuit to ground, faulty sensor.	Check wiring and connections. Replace sensor if required.				
60	01	FUEL PRESSURE - RAIL PRESSURE SENSOR OFFSET	EXCEEDED UPPER LIMIT		Faulty rail pressure sensor.	Replace sensor.				

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
60	02	FUEL PRESSURE - RAIL PRESSURE SENSOR OFFSET	BELOW LOWER LIMIT		Faulty rail pressure sensor.	Replace sensor.				
62	01	FUEL PRESSURE - FAULT ON THE FUEL DRUCK CONTROL OF THE RAIL (POSITIVE DEVIATION)	EXCEEDED UPPER LIMIT		High pressure circuit fuel leakage.	Check fuel feed system.				Fuel management and pressure failure in rail.
62	01	FUEL PRESSURE - FAULT ON THE FUEL DRUCK CONTROL OF THE RAIL (POSITIVE DEVIATION)	EXCEEDED UPPER LIMIT		Injector jammed in fuel passage open position.	Check hydraulic and mechanical efficiency of injectors.				Fuel management and pressure failure in rail.
62	01	FUEL PRESSURE - FAULT ON THE FUEL DRUCK CONTROL OF THE RAIL (POSITIVE DEVIATION)	EXCEEDED UPPER LIMIT		MPROP adjuster open movement jammed.	Check efficiency of MPROP adjuster.				Fuel management and pressure failure in rail.
62	01	FUEL PRESSURE - FAULT ON THE FUEL DRUCK CONTROL OF THE RAIL (POSITIVE DEVIATION)	EXCEEDED UPPER LIMIT		Faulty high pressure pump.	Check efficiency of high pressure pump.				Fuel management and pressure failure in rail.

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
63	01	FUEL PRESSURE - FAULT ON THE FUEL DRUCK CONTROL OF THE RAIL (NEGATIVE DEVIATION)	EXCEEDED UPPER LIMIT		MPROP adjuster open movement jammed.	Check efficiency of MPROP adjuster.				Fuel management and pressure failure in rail.
64	01	FUEL PRESSURE - RAIL PRESSURE ERROR: TOO LOW	EXCEEDED UPPER LIMIT		High pressure circuit fuel leakage.	Check high pressure system. Replace high pressure pump if required.				Fuel management and pressure failure in rail.
65	01	FUEL PRESSURE - RAIL PRESSURE ERROR: TOO HIGH	EXCEEDED UPPER LIMIT		MPROP regulator jammed.	Check MPROP regulator, replace if required.				
66	01	FUEL PRESSURE - ERROR ON THE RAIL PRESSURE (EXCESSIVE DUTY CYCLE)	EXCEEDED UPPER LIMIT	Negative vehicle reaction with smoke in exhaust during acceleration.	High pressure circuit fuel leakage.	Check fuel feed system, replace high pressure pump if required. Faulty fuel feed system (fuel pump and filter jammed).				
67	01	FUEL PRESSURE - ERROR ON THE RAIL PRESSURE (EXCESSIVE)	EXCEEDED UPPER LIMIT	Engine off.	MPROP regulator jammed.	Check MPROP regulator, replace if required.				Replace pressure relief valve.

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
68	02	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	BELOW LOWER LIMIT			Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
68	04	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	NO SIGNAL		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
68	08	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
69	01	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY	EXCEEDED UPPER LIMIT	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.
69	02	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY	BELOW LOWER LIMIT	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.
6A	01	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY	EXCEEDED UPPER LIMIT	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
6A	02	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY	BELOW LOWER LIMIT	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.
6B	01	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY	EXCEEDED UPPER LIMIT	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.
6B	02	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY	BELOW LOWER LIMIT	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.
6C	01	VEHICLE - EDC LAMP	EXCEEDED UPPER LIMIT	Warning light not working.	Short-circuit to positive.	Check correct operation of warning light using "Active diagnostic" procedure.				Warning light should come on for approximately 5 seconds at key-on. Check wiring and connections if this does not occur.

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
6C	02	VEHICLE - EDC LAMP	BELOW LOWER LIMIT	Warning light not working.	Short-circuit to ground.	Check correct operation of warning light using "Active diagnostic" procedure.				Warning light should come on for approximately 5 seconds at key-on. Check wiring and connections if this does not occur.
6C	04	VEHICLE - EDC LAMP	NO SIGNAL	Warning light not working.	Open circuit, bulb disconnected.	Check correct operation of warning light using "Active diagnostic" procedure.				Warning light should come on for approximately 5 seconds at key-on. Check wiring and connections if this does not occur.
6C	08	VEHICLE - EDC LAMP	SIGNAL NOT PLAUSIBLE	Warning light not working.	Wiring problems.	Check wiring and connections. Replace sensor if required.				Warning light should come on for approximately 5 seconds at key-on. Check wiring and connections if this does not occur.
6D	08	ENGINE 2 - INTERNAL ECU FAULT (PLAUSIBILITY ERROR +15)	SIGNAL NOT PLAUSIBLE			Check wiring and connections.				Key 15 off during initialisation.
6E	08	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
75	01	VEHICLE SPEED SENSOR SIGNAL	- EXCEEDED UPPER LIMIT	Speed of 170 km/h exceeded.		Check correct calibration of speedometer.				Encourage driver to use the vehicle correctly.
75	04	VEHICLE SPEED SENSOR SIGNAL	- NO SIGNAL		Interrupted wiring between vehicle speed sensor and instrument panel.	Check wiring and connections between vehicle speed sensor and instrument panel.				
75	04	VEHICLE SPEED SENSOR SIGNAL	- NO SIGNAL		Wiring interrupted between instrument panel and EDC ECU.	Check wiring and connections between instrument panel and EDC ECU.				Intervention required if instrument panel indicates vehicle speed.
75	04	VEHICLE SPEED SENSOR SIGNAL	- NO SIGNAL		Vehicle speed sensor disconnected or failed.	Check correct assembly and efficiency of vehicle speed sensor.				
75	08	VEHICLE SPEED SENSOR SIGNAL	- SIGNAL NOT PLAUSIBLE		Vehicle speed sensor disconnected or failed.	Check correct assembly and efficiency of vehicle speed sensor.				
75	08	VEHICLE SPEED SENSOR SIGNAL	- SIGNAL NOT PLAUSIBLE	Vehicle speed on instrument panel does not increase sensibly.	Wrong speedometer setting.	Check correct calibration of speedometer.				
77	01	VEHICLE SPEED SENSOR SIGNAL	- EXCEEDED UPPER LIMIT	Wrong vehicle speed indication.	Wrong speedometer setting.	Check correct calibration of speedometer.				
77	02	VEHICLE SPEED SENSOR SIGNAL	- BELOW LOWER LIMIT	Wrong vehicle speed indication.	Wrong speedometer setting.	Check correct calibration of speedometer.				

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
77	08	VEHICLE - VEHICLE SPEED SENSOR / SIGNAL	SIGNAL NOT PLAUSIBLE	Wrong vehicle speed indication.	Wrong speedometer setting.	Check correct calibration of speedometer.				
79	08	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

2nd Section

SYMPTOMS

The second section describes possible trouble that is **not identifiable by the control unit and is**

SPECIFIC TO THE COMMON RAIL SYSTEM AND THE NEW HW ENGINE

HYDRAULIC

ELECTRIC

MECHANICAL

other than conventional defects

(the aim is to guide the diagnostic approach to a new system, not to restate basic concepts that are considered to have already been acquired by the repairer).



The possible trouble already identified by the control unit, described in the 1st Section, is not repeated here (e.g., the engine cuts out as a result of defect 8.1).

If there are errors stored in the control unit memory, refer to the 1st troubleshooting section.

- ☐ The engine cuts out or fails to start.
- ☐ The engine fails to start (considerable exhaust smoke).
- ☐ The engine starts with difficulty.
- ☐ The engine fails to reach its top performance.

SYMPTOM	SYSTEM REACTION	POSSIBLE CAUSE	TESTS OR RECOMMENDED ACTION	NOTES
The engine cuts out or fails to start.	The EDC indicator light fails to come on. The starter motor turns but the engine fails to start.	EDC control unit not powered: fuse blown.	Check fuse No. 23 (25A). If the fuse has blown, find and eliminate the cause of the overload before replacing it.	
The engine cuts out or fails to start.	The EDC indicator light fails to come on. The starter motor turns but the engine fails to start.	EDC control unit not powered: the main relay is not powered.	Check the wiring upstream from the main relay to find any break in the circuit.	
The engine cuts out or fails to start.	The EDC control unit is powered, the starter motor turns but the engine fails to start. The rail pressure does NOT reach 200 bar.	Air intake in the supply circuit between the tank and motor pump.	Check the integrity of the pipe and check that the quick couplings on the CILC (fuel level indicator assembly) and on the motor pump inlet are fitted properly. Replace any non-conforming parts.	
The engine cuts out or fails to start.	The EDC control unit is powered, the starter motor turns but the engine fails to start. The rail pressure does NOT reach 200 bar.	Pre-filter clogged.	Inspect and replace the pre-filter if any debris is found inside.	The pre-filter is transparent and any debris is easy to see.
The engine cuts out or fails to start.	The EDC control unit is powered, the starter motor turns but the engine fails to start. The rail pressure does NOT reach 200 bar.	Low-pressure pipe between motor pump and high-pressure pump inlet choked or with large leak.	Inspect the pipe and replace the relevant section.	
The engine cuts out or fails to start.	The EDC control unit is powered, the starter motor turns but the engine fails to start. The rail pressure does NOT reach 200 bar.	Fuel filter greatly clogged (within certain limits it only involves difficult starting).	Replace the filter.	If the filter clogging indicator system has not worked, check the relevant electric circuit and restore its operation.

SYMPTOM	SYSTEM REACTION	POSSIBLE CAUSE	TESTS OR RECOMMENDED ACTION	NOTES
The engine cuts out or fails to start.	The EDC control unit is powered, the starter motor turns but the engine fails to start. The rail pressure does NOT reach 200 bar.	Rail pressure relief valve jammed open or lost its setting (continually discharges towards the tank).	If fuel has come out of the valve exhaust pipe while driving with the starter motor, change the valve.	
The engine cuts out or fails to start.	The EDC control unit is powered, the starter motor turns but the engine fails to start. The rail pressure does NOT reach 200 bar.	Mechanical defect in the gear pump, pressure regulator and the pumping elements of the high-pressure pump.	After checking there is fuel in the tank and excluding every other possibility (see 1 st Troubleshooting Section), replace the high-pressure pump together with the pressure regulator.	
The engine cuts out or fails to start.	The starter motor turns but the engine fails to start. The rail pressure during starting regularly rises above 200 bar.	EGR pneumatic valve jammed open and air throttle valve jammed shut.	Check and replace the defective components.	
The engine starts with difficulty.	The EDC control unit is powered, the starter motor turns but the engine starts only after insisting a long time. Very slow increase in rail pressure.	The fuel motor pump is not powered (no buzzing is heard with the key ON for 9 sec.).	Check that no electric cable has disconnected from the motor pump. Check the wiring between the control relay and the motor pump to identify any break in the circuit.	After starting, with a load request the engine goes into recovery (if due to insufficient fuel reaching the high-pressure pump error 8.1 is detected, see 1 st Section).
The engine starts with difficulty.	The rail pressure during starting regularly rises above 200 bar.	Injector mechanically jammed shut.	Perform the Engine Test (cylinder efficiency) to identify the defective injector and replace it.	Depending on the extent of the jamming, the control unit might detect a lack of balance between the cylinders (See error 3.1 – 3.2 – 3.3 – 3.4, 1 st Section).
The engine starts with difficulty.	The rail pressure during starting regularly rises above 200 bar.	EGR pneumatic valve jammed open or air throttle valve mechanically jammed shut.	Check which component is defective and replace it.	

SYMPTOM	SYSTEM REACTION	POSSIBLE CAUSE	TESTS OR RECOMMENDED ACTION	NOTES
The engine starts with difficulty.	The rail pressure during starting does not reach 200 bar immediately.	Air intake in the supply circuit between the tank and motor pump.	Check the integrity of the pipe and check that the quick couplings on the CILC (fuel level indicator assembly) and on the motor pump inlet are fitted properly. Replace any non-conforming parts.	
The engine starts with difficulty.	The rail pressure during starting does not reach 200 bar immediately.	The motor pump is not powered (no buzzing is heard with the key ON for 9 sec.).	Check the wiring between the control relay and the motor pump.	
The engine starts with difficulty.	The rail pressure during cranking does not reach 200 bar immediately.	Low-pressure pipe choked or broken or leaking.	Inspect the pipe and replace the relevant section.	
The engine starts with difficulty.	The rail pressure during cranking does not reach 200 bar immediately.	Fuel filter very clogged.	Replace the filter. If the filter clogging indicator system has not worked, check the relevant circuit and restore its operation.	
The engine fails to reach top performance	(with no derating implemented by the control unit)	Throttle pedal potentiometer does not go to the end of its travel.	Read parameters, check the signal reaches 100%. If it does not, check the physical integrity of the potentiometer and replace it if necessary.	If there are errors saved in the control unit memory, refer to the 1 st Troubleshooting Section.
The engine fails to reach top performance	(with no derating implemented by the control unit)	EGR pneumatic valve jammed open or throttle valve jammed shut.	Check which is the defective component and replace it.	
The engine fails to reach top performance	(with no derating implemented by the control unit)	Injector jammed shut.	Find the defective injector (cylinder efficiency test with the diagnostic instrument) and replace it.	

SYMPTOM	SYSTEM REACTION	POSSIBLE CAUSE	TESTS OR RECOMMENDED ACTION	NOTES
The engine fails to reach top performance	(with no derating implemented by the control unit)	Fuel filter greatly clogged.	Change the filter. If the filter clogging indicator system has not worked, check the relevant circuit and restore its operation.	
The engine fails to reach top performance	(with no derating implemented by the control unit)	The motor pump is not powered (no buzzing is heard with the key ON for 9 sec.).	Check the wiring between the control relay and the motor pump.	

Ist Section
EDT-FMI error codes
with EDC 16 central unit
version software P315 V4b

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
01	01	19	VEHICLE - AIR-CONDITIONER COMPRESSOR RELAY (DTC 1)	EXCEED-ING NOR-MAL RANGE	A/C compressor always on.	Short-circuit to positive.	Check wiring and connections. Replace relay if required.				
01	02	19	VEHICLE - AIR-CONDITIONER COMPRESSOR RELAY (DTC 1)	EXCEED-ING NOR-MAL RANGE	Check correct operation of warning light using "Active diagnostic" procedure.	Short-circuit to ground.	Check wiring and connections. Replace relay if required.				
01	04	19	VEHICLE - AIR-CONDITIONER COMPRESSOR RELAY (DTC 1)	NO SIGNAL	A/C compressor not working.	Open circuit, relay disconnected.	Check wiring and connections. Replace relay if required.				
01	08	19	VEHICLE - AIR-CONDITIONER COMPRESSOR RELAY (DTC 1)	SIGNAL NOT PLAUS-IBLE	A/C compressor not working.	Open circuit, relay disconnected.	Check wiring and connections. Replace relay if required.				
02	04	19	VEHICLE - AIR-CONDITIONER COMPRESSOR RELAY (DTC 2)	NO SIGNAL	A/C compressor not working.	No CAN line signal.	Check wiring and connections. Replace relay if required.				
02	08	19	VEHICLE - AIR-CONDITIONER COMPRESSOR RELAY (DTC 2)	SIGNAL NOT PLAUS-IBLE	A/C compressor not working.	Non plausible CAN line signal.	Check wiring and connections. Replace relay if required.				
03	01	97	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 3)	EXCEED-ING NOR-MAL RANGE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
03	02	97	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 3)	EXCEEDING NORMAL RANGE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
03	04	97	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 3)	NO SIGNAL		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
03	08	97	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 3)	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
05	01	87	EGR - INCORRECT DEBIT-METER SIGNAL	EXCEEDING NORMAL RANGE	Possible poor performance in acceleration.	Flow meter short-circuit or open circuit.	Check integrity of flow meter and wiring between flow meter connector and EDC connector.				Power reduction and deactivated EGR function.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
05	02	87	EGR - INCORRECT DEBI-METER SIGNAL	EXCEED-ING NOR-MAL RANGE	Possible poor performance in acceleration.	Flow meter short-circuit or open circuit.	Check integrity of flow meter and wiring between flow meter connector and EDC connector.				Power reduction and deactivated EGR function.
07	01	87	EGR - DEBI-METER SIGNAL OUT OF LIMIT	EXCEED-ING NOR-MAL RANGE	Possible poor performance in acceleration.	Flow meter short-circuit or open circuit.	Check integrity of flow meter and wiring between flow meter connector and EDC connector.				Power reduction and deactivated EGR function.
07	02	87	EGR - DEBI-METER SIGNAL OUT OF LIMIT	EXCEED-ING NOR-MAL RANGE	Possible poor performance in acceleration.	Flow meter short-circuit or open circuit.	Check integrity of flow meter and wiring between flow meter connector and EDC connector.				Power reduction and deactivated EGR function.
08	01	14	VEHICLE - ACCELERATOR PEDAL I	EXCEED-ING NOR-MAL RANGE	Incorrect accelerator pedal operation, engine idling at 1500 rpm.	Short-circuit to positive, voltage exceeding 4700 mV.	In "Measurable parameters" environment, check that the "accelerator pedal position" parameter changes proportionally to the position of the pedal from 0% to 100%. Check wiring and connections. Replace accelerator pedal if required.				Make sure that accelerator pedal travel is not hindered.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
08	02	14	VEHICLE - ACCELERATOR PEDAL I	EXCEEDING NORMAL RANGE	Incorrect accelerator pedal operation, engine idling at 1500 rpm.	Short-circuit to ground.	In "Measurable parameters" environment, check that the "accelerator pedal position" parameter changes proportionally to the position of the pedal from 0% to 100%. Check wiring and connections. Replace accelerator pedal if required.				Make sure that accelerator pedal travel is not hindered.
08	08	14	VEHICLE - ACCELERATOR PEDAL I	SIGNAL NOT PLAUSIBLE	Incorrect accelerator pedal operation, engine idling at 1500 rpm.	Accelerator pedal potentiometer 1 and accelerator pedal potentiometer 2 values not plausible.	In "Measurable parameters" environment, check that the "accelerator pedal position" parameter changes proportionally to the position of the pedal from 0% to 100%. Check wiring and connections. Replace accelerator pedal if required.				Make sure that accelerator pedal travel is not hindered.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
09	01	14	VEHICLE - ACCELERATOR PEDAL 2	EXCEEDING NORMAL RANGE	Incorrect accelerator pedal operation, engine idling at 1500 rpm.	Short-circuit to positive, voltage exceeding 4700 mV.	In "Measurable parameters" environment, check that the "accelerator pedal position" parameter changes proportionally to the position of the pedal from 0% to 100%. Check wiring and connections. Replace accelerator pedal if required.				Make sure that accelerator pedal travel is not hindered.
09	02	14	VEHICLE - ACCELERATOR PEDAL 2	EXCEEDING NORMAL RANGE	Incorrect accelerator pedal operation, engine idling at 1500 rpm.	Short-circuit to ground.	In "Measurable parameters" environment, check that the "accelerator pedal position" parameter changes proportionally to the position of the pedal from 0% to 100%. Check wiring and connections. Replace accelerator pedal if required.				Make sure that accelerator pedal travel is not hindered.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
09	08	14	VEHICLE - ACCELERATOR PEDAL 2	SIGNAL NOT PLAUSIBLE	Incorrect accelerator pedal operation, engine idling at 1500 rpm.	Accelerator pedal potentiometer 1 and accelerator pedal potentiometer 2 values not plausible.	In "Measurable parameters" environment, check that the "accelerator pedal position" parameter changes proportionally to the position of the pedal from 0% to 100%. Check wiring and connections. Replace accelerator pedal if required.				Make sure that accelerator pedal travel is not hindered.
0A	01	25	ENGINE I - ATMOSPHERIC PRESSURE SENSOR	EXCEEDING NORMAL RANGE	Possible smokiness at high altitude. Problematic cranking at high altitude.	Faulty environmental pressure sensor in ECU.	Replace ECU.				
0A	02	25	ENGINE I - ATMOSPHERIC PRESSURE SENSOR	EXCEEDING NORMAL RANGE	Possible smokiness at high altitude. Problematic cranking at high altitude.	Faulty environmental pressure sensor in ECU.	Replace ECU.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
0B	08	17	VEHICLE - ACCELERATOR PEDAL/BRAKE PEDAL SUSPECT	SIGNAL NOT PLAUSIBLE	Power reduction: engine rpm drop to idling speed.	Press brake and accelerator at the same time.	Check wiring and connections. Replace sensor if required.				If the brake is actuated while the accelerator is pressed, the motor runs on slow idling until the brake is released, so that the vehicle can be stopped even if the pedal of the accelerator remains stuck on the intermediate position. Instead you can accelerate while the pedal is pressed without the interference of safety measures.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
0C	01	44	EGR - AIR MASS SUPPLY TOO LOW	EXCEED-ING NORMAL RANGE	EGR off. Emissions not compliant with law. Derated performance and smoke at high engine rpm.	EGR monitoring: incorrect EGR percentage actuation calculated by ECU.	Check, if the EGR pneumatic valve is not locked in Open or Closed-Position 2) Check, that the solenoid valve and the EGR pneumatic valve is not squashed or holed or detached 3) Check the EGR solenoid valve-functionality 4) Check the solenoid valve-integrity by means of a multimeter 5) Check the wiring harness between the solenoid valve and the EDC connector.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
0D	02	44	EGR - AIR MASS SUPPLY TOO HIGH	EXCEED-ING NORMAL RANGE	EGR off. Emissions not compliant with law. Derated performance and smoke at high engine rpm.	EGR monitoring: incorrect EGR percentage actuation calculated by ECU.	Check, if the EGR pneumatic valve is not locked in Open or Closed-Position 2) Check, that the solenoid valve and the EGR pneumatic valve is not squashed or holed or detached 3) Check the EGR solenoid valve-functionality 4) Check the solenoid valve-integrity by means of a multimeter 5) Check the wiring harness between the solenoid valve and the EDC connector.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
11	01	24	ENGINE 1 - BOOST PRESSURE SENSOR	EXCEEDING NORMAL RANGE	Positive power reduction and smoke in exhaust.		Check wiring and connections. Possibly replace sensor. Check in "measurable parameter" environment that atmospheric pressure sensor and turbo charger air pressure sensor values are similar when engine is off.				Possible smoke in exhaust during acceleration. Replace if required.
11	02	24	ENGINE 1 - BOOST PRESSURE SENSOR	EXCEEDING NORMAL RANGE	Positive power reduction and smoke in exhaust.		Check wiring and connections. Replace sensor if required.				Possible smoke in exhaust during acceleration. Replace if required.
11	08	24	ENGINE 1 - BOOST PRESSURE SENSOR	SIGNAL NOT PLAUSIBLE	Positive power reduction and smoke in exhaust.	Faulty sensor.	Check wiring and connections. Replace sensor if required.				
12	01	35	ENGINE 2 - BATTERY VOLTAGE	EXCEEDING NORMAL RANGE	Problematic cranking.	Flat battery, interrupted wiring.	Check battery state with diagnostic tool (measurable parameters). Check wiring and connections.				Replace alternator, regulator or battery.
12	02	35	ENGINE 2 - BATTERY VOLTAGE	EXCEEDING NORMAL RANGE	Engine does not start.	Faulty battery, faulty alternator, faulty ECU.	Check with diagnostic tool.				Replace battery, alternator or ECU if required.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
13	08	16	VEHICLE - BRAKE PEDAL SIGNAL ERROR	SIGNAL NOT PLAUSIBLE	Brake signal plausibility, possibly no brake lights, Cruise Control / PTO not working.	The two switch states are different.	Check wiring and connections. Replace sensor if required.				
14	01	21	ENGINE - COOLANT TEMPERATURE SENSOR	EXCEEDING NORMAL RANGE	Problematic cold cranking. Possible power reduction.	Faulty sensor, interrupted wiring.	Check wiring and connections. Replace sensor if required.	1- Measure type: Resistance (KOhm) Measure point 1: Sensor Pin: 1 Measure point 2: Sensor Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A58 Measure point 2: Sensor Pin: 1 3- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A41 Measure point 2: Sensor Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Min. value: 0,11 KOhm; Max. value: 48,3 KOhm; Typical Value: 2,5 KOhm; 2- Typical Value: 0,1 Ohm; 3- Typical Value: 0,1 Ohm;	

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
14	02	21	ENGINE 1 - COOLANT TEMPERATURE SENSOR	EXCEED-ING NORMAL RANGE	Problematic cold cranking. Possible power reduction.	Faulty sensor, interrupted wiring.	Check wiring and connections. Replace sensor if required.	1- Measure type: Resistance (KOhm) Measure point 1: Sensor Pin: 1 Measure point 2: Sensor Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A58 Measure point 2: Sensor Pin: 1 3- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A41 Measure point 2: Sensor Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Min. value: 0,11 KOhm; Max. value: 48,3 KOhm; Typical Value: 2,5 KOhm; 2- Typical Value: 0,1 Ohm; 3- Typical Value: 0,1 Ohm;	

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
14	08	21	ENGINE COOLANT TEMPERATURE SENSOR	SIGNAL NOT PLAUSIBLE	Problematic cold cranking. Possible power reduction.	Faulty sensor, interrupted wiring.	Check wiring and connections. Replace sensor if required.	1- Measure type: Resistance (KOhm) Measure point 1: Sensor Pin: 1 Measure point 2: Sensor Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A58 Measure point 2: Sensor Pin: 1 3- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A41 Measure point 2: Sensor Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Min. value: 0,11 KOhm; Max. value: 48,3 KOhm; Typical Value: 2,5 KOhm; 2- Typical Value: 0,1 Ohm; 3- Typical Value: 0,1 Ohm;	

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
15	01	21	ENGINE COOLANT TEMPERATURE SENSOR (TEST)	EXCEED-ING NORMAL RANGE		Faulty coolant temperature sensor.	Replace sensor.	1- Measure type: Resistance (KOhm) Measure point 1: Sensor Pin: 1 Measure point 2: Sensor Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A58 Measure point 2: Sensor Pin: 1 3- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A41 Measure point 2: Sensor Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Min. value: 0,11 KOhm; Max. value: 48,3 KOhm; Typical Value: 2,5 KOhm; 2- Typical Value: 0,1 Ohm; 3- Typical Value: 0,1 Ohm;	
IE	08	15	VEHICLE CLUTCH SIGNAL SUSPECT	SIGNAL NOT PLAUSIBLE	Clutch switch: signal either not plausible or not present. Cruise Control / PTO not working or engine revs up to maximum speed when clutch pedal is pressed and Cruise control / PTO is on.	Gear shift detected without pressing brake pedal.	Check wiring and connections. Replace sensor if required.				The anomaly caused by incomplete clutch operation if everything is OK.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
20	01	86	EGR - EGR POWER - ST. SHORT TO BATT.	EXCEED-ING NOR-MAL RANGE		EGR solenoid valve short-circuit to battery.	1) Check integrity of solenoid valve with multi-meter. 2) Check wiring between solenoid valve and EDC connector.				EGR either not working or always working. Emissions not compliant with law. No reaction perceivable by driver.
21	02	86	EGR - SHORT CIRCUIT TO GROUND ON EGR VALVE	EXCEED-ING NOR-MAL RANGE		Solenoid valve short-circuit to ground.	1) Check integrity of solenoid valve with multi-meter. 2) Check wiring between solenoid valve and EDC connector.				EGR either not working or always working. Emissions not compliant with law. No reaction perceivable by driver.
22	04	86	EGR - OPEN CIRCUIT ON EGR VALVE	NO SIG-NAL		EGR solenoid valve short-circuit or open circuit.	1) Check integrity of solenoid valve with multi-meter. 2) Check wiring between solenoid valve and EDC connector.				EGR either not working or always working. Emissions not compliant with law. No reaction perceivable by driver.
22	08	86	EGR - OPEN CIRCUIT ON EGR VALVE	SIGNAL NOT PLAUS-IBLE		EGR solenoid valve short-circuit or open circuit.	1) Check integrity of solenoid valve with multi-meter. 2) Check wiring between solenoid valve and EDC connector.				EGR either not working or always working. Emissions not compliant with law. No reaction perceivable by driver.
24	01	62	ENGINE SPEED - CAMSHAFT SENSOR	EXCEED-ING NOR-MAL RANGE	Possible prob-lematic cranking.	No signal, open circuit.	Check wiring and connec-tions.				Flywheel sensor timing signal adopted if cam-shaft signal is not correct.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
24	02	62	ENGINE SPEED - CAMSHAFT SENSOR	EXCEED-ING NORMAL RANGE	Possible problematic cold cranking.	No signal, open circuit, faulty sensor.	Check correct assembly of sensor and phonic wheel, check engine timing.				Flywheel sensor timing signal adopted if camshaft signal is not correct.
25	01	61	ENGINE SPEED - CRANKSHAFT SENSOR	EXCEED-ING NORMAL RANGE	Problematic cold cranking, power reduction (possible noise due to missed pre-injection).	Faulty sensor.	Check wiring and connections.				Camshaft sensor speed adopted if signal is not present.
25	02	61	ENGINE SPEED - CRANKSHAFT SENSOR	EXCEED-ING NORMAL RANGE	Problematic cold cranking, power reduction (possible noise due to missed pre-injection).	Faulty sensor.	Check wiring and connections.				Camshaft sensor speed adopted if signal is not present.
26	01	63	ENGINE SPEED - FAULT BETWEEN FLY-WHEEL SENSOR AND CAM-SHAFT	EXCEED-ING NORMAL RANGE	Possible power reduction.	Incorrect camshaft phonic wheel assembly.	Check wiring, connections and sensor, check that phonic wheel is fitted correctly.				Longer cranking time.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
28	01	23	ENGINE I - FUEL TEMPERATURE SENSOR	EXCEED- ING NOR- MAL RANGE	Possible power reduction.	Short-circuit to positive, excessively low temperature is detected.	Check wiring and connections. Replace sensor if required.	1- Measure type: Resistance (Ohm) Measure point 1: Sensor Pin: 1 Measure point 2: Sensor Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A52 Measure point 2: Sensor Pin: 1 3- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A51 Measure point 2: Sensor Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Typical Value: 0,1 Ohm;	

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
28	02	23	ENGINE I - FUEL TEMPERATURE SENSOR	EXCEED-ING NOR-MAL RANGE	Possible power reduction.	Short-circuit to ground, excessively high temperature is detected.	Check wiring and connections. Replace sensor if required.	1- Measure type: Resistance (Ohm) Measure point 1: Sensor Pin: 1 Measure point 2: Sensor Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A52 Measure point 2: Sensor Pin: 1 3- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A51 Measure point 2: Sensor Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Typical Value: 0,1 Ohm;	
29	01	29	ENGINE I - FAN RELAY	EXCEED-ING NOR-MAL RANGE	Fan relay not working.	Fan relay short-circuit to positive.	Check wiring and connections. Replace relay if required.				Possible increased fuel consumption. Possible engine overheating and power reduction.
29	02	29	ENGINE I - FAN RELAY	EXCEED-ING NOR-MAL RANGE	Fan relay not working.	Fan relay short-circuit to ground.	Check wiring and connections. Replace relay if required.				Possible increased fuel consumption. Possible engine overheating and power reduction.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
29	04	29	ENGINE I - FAN RELAY	NO SIGNAL	Fan relay not working.		Check wiring and connections. Replace relay if required.				Possible increased fuel consumption. Possible engine overheating and power reduction.
29	08	29	ENGINE I - FAN RELAY	SIGNAL NOT PLAUSIBLE	Fan relay not working.		Check wiring and connections. Replace relay if required.				Possible increased fuel consumption. Possible engine overheating and power reduction.
2A	01	28	ENGINE I - PRE-HEATING RELAY FUEL FILTER	EXCEEDING NORMAL RANGE	Fuel filter pre-heater relay not working.	Filter heater relay short-circuit to positive - Heater always on also at fuel temperature > 5° C.	Check wiring and connections. Replace relay if required.				Battery goes flat.
2A	02	28	ENGINE I - PRE-HEATING RELAY FUEL FILTER	EXCEEDING NORMAL RANGE	Fuel filter pre-heater relay not working.	Filter heater relay short-circuit to ground.	Check wiring and connections. Replace relay if required.				Battery goes flat.
2A	04	28	ENGINE I - PRE-HEATING RELAY FUEL FILTER	NO SIGNAL	Fuel filter pre-heater relay not working.		Check wiring and connections. Replace relay if required.				Battery goes flat.
2A	08	28	ENGINE I - PRE-HEATING RELAY FUEL FILTER	SIGNAL NOT PLAUSIBLE	Fuel filter pre-heater relay not working.		Check wiring and connections. Replace relay if required.				Battery goes flat.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
2F	01	37	ENGINE GLOW RELAY 2 - PLUGS	EXCEED-ING NOR-MAL RANGE	Possible prob-lematic cold cranking.	Short-circuit to positive, glow plugs always on also with ECU off, possible battery deployment.	Check wiring and connections. Replace relay if required.				
2F	02	37	ENGINE GLOW RELAY 2 - PLUGS	EXCEED-ING NOR-MAL RANGE		Short-circuit to ground, glow plugs always on.	Check wiring and connections. Replace relay if required.				
2F	04	37	ENGINE GLOW RELAY 2 - PLUGS	NO SIG-NAL	Possible prob-lematic cold cranking.	Faulty wiring.	Check wiring and connections. Replace relay if required.				Faulty diagnostic light.
2F	08	37	ENGINE GLOW RELAY 2 - PLUGS	SIGNAL NOT PLAUS-IBLE	Possible prob-lematic cold cranking.	Faulty relay, wiring interrupted.	Check wiring and connections. Replace relay if required.				Possible increased fuel consumption. Possible engine overheating and power reduction.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
30	01	36	ENGINE 2 - GLOW PLUG W/LIGHT	EXCEED-ING NORMAL RANGE	Warning light always off. Problematic cold cranking. Pre-heater warning light always on.	Short-circuit to positive.	Check wiring and connections. Replace sensor if required.				The driver does not wait pre-heating even when the room temperatures are low, because no warning light signal is enabled. Preheating works, but with cold start-up no indication is available that tells you when to start the motor because the light is always turned on.
30	02	36	ENGINE 2 - GLOW PLUG W/LIGHT	EXCEED-ING NORMAL RANGE	Warning light always off. Problematic cold cranking. Pre-heater warning light always on.	Short-circuit to ground.	Check wiring and connections. Replace sensor if required.				The driver does not wait pre-heating even when the room temperatures are low, because no warning light signal is enabled. Preheating works, but with cold start-up no indication is available that tells you when to start the motor because the light is always turned on.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
30	04	36	ENGINE 2 - GLOW PLUG W/LIGHT	NO SIGNAL	Warning light always off. Problematic cold cranking. Pre-heater warning light always on.		Check wiring and connections. Replace sensor if required.				Warning light off during pre-heating. Replace bulb if required.
30	08	36	ENGINE 2 - GLOW PLUG W/LIGHT	SIGNAL NOT PLAUSIBLE	Warning light always off. Problematic cold cranking. Pre-heater warning light always on.		Check wiring and connections. Replace sensor if required.				Warning light off during pre-heating. Replace bulb if required.
31	01	39	ENGINE 2 - GLOW PLUGS	EXCEEDING NORMAL RANGE	Possible problematic cold cranking.	Short-circuit to positive.	Check wiring and connections. Check electrical system between relay and glow plugs.				Relay unit always on also with ECU off, possible battery deployment.
32	01	91	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 50)	EXCEEDING NORMAL RANGE		Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
33	01	92	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 51)	EXCEED-ING NORMAL RANGE	The engine switching off-data are not memorized. The failures memory is lost, only the present failures and not the intermittent ones can be read, the idling speed, which can be eventually set by the Cruise Control commands, remains not memorized.	Faulty ECU EE-PROM.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
33	02	92	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 51)	EXCEED-ING NORMAL RANGE	The engine switching off-data are not memorized. The failures memory is lost, only the present failures and not the intermittent ones can be read, the idling speed, which can be eventually set by the Cruise Control commands, remains not memorized.	Faulty ECU EE-PROM.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
33	04	92	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 51)	NO SIGNAL	The engine switching off-data are not memorized. The failures memory is lost, only the present failures and not the intermittent ones can be read, the idling speed, which can be eventually set by the Cruise Control commands, remains not memorized.	Faulty ECU EEPROM.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
33	08	92	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 51)	SIGNAL NOT PLAUSIBLE	The engine switching off-data are not memorized. The failures memory is lost, only the present failures and not the intermittent ones can be read, the idling speed, which can be eventually set by the Cruise Control commands, remains not memorized.	Faulty ECU EEPROM.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
34	08	91	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 52)	SIGNAL NOT PLAUSIBLE	There have been cases in which this fault is stored by the E.C.U. following after failures in engine speed measurement. Failure is usually associated to sudden ignition missing at >3000 rpm engine speed, with engine disconnection following. Reading failure memory, fault code DTC52 is displayed, sometimes followed by DTC56 (relating to fault on Immobilizer).	The fault is attributable to the gear wheel, which probably presents incorrect tooth shape.	Check there are no mechanical faults on the teeth of the gear wheel fitted to the engine drive shaft.				
35	08	91	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 53)	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
36	08	91	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 54)	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
37	01	91	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 55)	EXCEEDING NORMAL RANGE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
38	02	91	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 56)	EXCEEDING NORMAL RANGE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
39	01	22	ENGINE I - AIR TEMPERATURE SENSOR	EXCEEDING NORMAL RANGE	Problematic cranking, smoke, problematic acceleration.		Check wiring and connections. Replace sensor if required.	Measure type: Resistance (KOhm) Measure point 1: Sensor Pin: 1 Measure point 2: Sensor Pin: 2	Connector Not connected; Key +15 OFF;	Typical Value: 2,5 KOhm;	Air temperature sensor and built-in pressure sensor. The sensor is fitted on flow meter in engines with EGR.
39	02	22	ENGINE I - AIR TEMPERATURE SENSOR	EXCEEDING NORMAL RANGE	Problematic cranking, smoke, problematic acceleration.	Short-circuit to ground, excessively high temperature is detected.	Check wiring and connections. Replace sensor if required.	Measure type: Resistance (KOhm) Measure point 1: Sensor Pin: 1 Measure point 2: Sensor Pin: 2	Connector Not connected; Key +15 OFF;	Typical Value: 2,5 KOhm;	Air temperature sensor and built-in pressure sensor. The sensor is fitted on flow meter in engines with EGR.
3A	02	93	ELECTRONIC CONTROL UNIT - IMMOBILISER	EXCEEDING NORMAL RANGE	The engine fails to start	Communication with Immobilizer ECU problems on CAN Line.	Check integrity of CAN Line, run Immobilizer ECU diagnostics and wait for indications provided.	Measure type: Resistance (Ohm) Measure point 1: Diagnostic socket. Pin: 21 Measure point 2: Diagnostic socket. Pin: 22	Connector Connected; Key +15 OFF;	Typical Value: 60 Ohm Ohm;	

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
3C	01	57	INJECTOR - BENCH I (DTC 60)	EXCEED-ING NORMAL RANGE	Engine not working properly, possible power reduction.	Injector wiring short-circuit.	Check wiring and connections. Replace injector if required.	1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A3 I Measure point 2: Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A I Measure point 2: Pin: I 3- Measure type: Resistance (Ohm) Measure point 1: Pin: I Measure point 2: Pin: 2 4- Measure type: Resistance (Ohm) Measure point 1: Pin: I Measure point 2: Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF; 4- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; 4- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	Only two cylinders running.
3C	02	57	INJECTOR - BENCH I (DTC 60)	EXCEED-ING NORMAL RANGE	Engine not working properly, possible power reduction.	Short-circuit to ground.	Check wiring and connections.				Only two cylinders running.
3C	08	57	INJECTOR - BENCH I (DTC 60)	SIGNAL NOT PLAUS-IBLE	Engine not working properly, possible power reduction.	Injector electrical system failure.	Check wiring and connections. Replace injector if required.				Only two cylinders running.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
3D	04	57	INJECTOR - BENCH 1 (DTC 61)	NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring disconnected.	Check wiring and connections. Replace injector if required.				Only two cylinders running.
3E	01	58	INJECTOR - BENCH 2 (DTC 62)	EXCEEDING NORMAL RANGE	Engine not working properly, possible power reduction.	Injector wiring short-circuit.	Check wiring and connections. Replace injector if required.	1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A1 Measure point 2: Pin: 1 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A31 Measure point 2: Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: Pin: 1 Measure point 2: Pin: 2 4- Measure type: Resistance (Ohm) Measure point 1: Injector 2 Pin: 1 Measure point 2: Injector 2 Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Connected; Key +15 OFF; 4- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm; 4- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	Only two cylinders running.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
3E	02	58	INJECTOR - BENCH 2 (DTC 62)	EXCEED-ING NORMAL RANGE	Engine not working properly, possible power reduction.	Short-circuit to ground.	Check wiring and connections.				Only two cylinders running.
3E	08	58	INJECTOR - BENCH 2 (DTC 62)	SIGNAL NOT PLAUS-IBLE	Engine not working properly, possible power reduction.	Injector electrical system failure.	Check wiring and connections. Replace injector if required.				Only two cylinders running.
3F	04	58	INJECTOR - BENCH 2 (DTC 63)	NO SIG-NAL	Engine not working properly, possible power reduction.	Injector wiring disconnected.	Check wiring and connections. Replace injector if required.				Only two cylinders running.
40	01	57	INJECTOR - STAGE A INJECTORS CHECK (INTERNAL ECU) (DTC 64)	EXCEED-ING NORMAL RANGE	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.				
40	02	57	INJECTOR - STAGE A INJECTORS CHECK (INTERNAL ECU) (DTC 64)	EXCEED-ING NORMAL RANGE	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
40	04	57	INJECTOR - STAGE A INJECTORS CHECK (INTERNAL ECU) (DTC 64)	NO SIGNAL	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.				
40	08	57	INJECTOR - STAGE A INJECTORS CHECK (INTERNAL ECU) (DTC 64)	SIGNAL NOT PLAUSIBLE	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.				
41	01	57	INJECTOR - STAGE B INJECTORS CHECK (INTERNAL ECU) (DTC 65)	EXCEEDING NORMAL RANGE	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
41	02	57	INJECTOR - STAGE B INJECTORS CHECK (INTERNAL ECU) (DTC 65)	EXCEED-ING NOR-MAL RANGE	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.				
41	04	57	INJECTOR - STAGE B INJECTORS CHECK (INTERNAL ECU) (DTC 65)	NO SIGNAL	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.				
41	08	57	INJECTOR - STAGE B INJECTORS CHECK (INTERNAL ECU) (DTC 65)	SIGNAL NOT PLAUS-IBLE	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.				
42	01	51	INJECTOR - INJECTOR I (DTC 66)	EXCEED-ING NOR-MAL RANGE							

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
42	01	53	INJECTOR - INJECTOR 1 (DTC 66)	EXCEEDING NORMAL RANGE	Engine not working properly, possible power reduction.	Short-circuit to positive.	Check wiring and connections. Replace injector if required.	1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A46 Measure point 2: Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A1 Measure point 2: Pin: 1 3- Measure type: Resistance (Ohm) Measure point 1: Pin: 1 Measure point 2: Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	Only three cylinders running.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
42	04	51	INJECTOR - INJECTOR I (DTC 66)	NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring short-circuit.	Check wiring and connections. Replace injector if required.	1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A1 Measure point 2: Pin: 1 2- Measure type: Resistance (Ohm) Measure point 1: Pin: 1 Measure point 2: Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A31 Measure point 2: Pin: 2 4- Measure type: Resistance (Ohm) Measure point 1: Pin: 1 Measure point 2: Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF; 4- Connector Connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm; 3- Typical Value: 0,1 Ohm; 4- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	Only two cylinders running.
42	04	51	INJECTOR - INJECTOR I (DTC 66)	NO SIGNAL							

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
42	08	51	INJECTOR - INJECTOR 1 (DTC 66)	SIGNAL NOT PLAUSIBLE	Engine not working properly, possible power reduction.	Injector not working properly.	Check wiring and connections. Replace injector if required.	1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A1 Measure point 2: Pin: 1 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A31 Measure point 2: Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: Pin: 1 Measure point 2: Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	Only three cylinders running.
42	08	51	INJECTOR - INJECTOR 1 (DTC 66)	SIGNAL NOT PLAUSIBLE							
43	04	51	INJECTOR - INJECTOR 1 (DTC 67)	NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring open circuit.	Check wiring and connections. Replace injector if required.				Only three cylinders running.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
44	01	54	INJECTOR - INJECTOR 2 (DTC 72)	EXCEED-ING NORMAL RANGE	Engine not working properly, possible power reduction.	Short-circuit to positive.	Check wiring and connections. Replace injector if required.	1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A46 Measure point 2: Injector 2 Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A1 Measure point 2: Injector 2 Pin: 1 3- Measure type: Resistance (Ohm) Measure point 1: Injector 2 Pin: 1 Measure point 2: Injector 2 Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	Only three cylinders running.
44	01	54	INJECTOR - INJECTOR 2 (DTC 72)	EXCEED-ING NORMAL RANGE							

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
44	04	54	INJECTOR - INJECTOR 2 (DTC 72)	NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring short-circuit.	Check wiring and connections. Replace injector if required.	1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A1 Measure point 2: Pin: 1 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A31 Measure point 2: Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: Pin: 1 Measure point 2: Pin: 2 4- Measure type: Resistance (Ohm) Measure point 1: Pin: 1 Measure point 2: Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Connected; Key +15 OFF; 4- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm; 4- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	Only two cylinders running.
44	04	54	INJECTOR - INJECTOR 2 (DTC 72)	NO SIGNAL							

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
44	08	54	INJECTOR - INJECTOR 2 (DTC 72)	SIGNAL NOT PLAUSIBLE	Engine not working properly, possible power reduction.	Injector not working properly.	Check wiring and connections. Replace injector if required.	1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A1 Measure point 2: Pin: 1 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A31 Measure point 2: Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: Pin: 1 Measure point 2: Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	Only three cylinders running.
44	08	54	INJECTOR - INJECTOR 2 (DTC 72)	SIGNAL NOT PLAUSIBLE							
45	04	54	INJECTOR - INJECTOR 2 (DTC 73)	NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring open circuit.	Check wiring and connections. Replace injector if required.				Only three cylinders running.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
46	01	52	INJECTOR - INJECTOR 3 (DTC 68)	EXCEEDING NORMAL RANGE	Engine not working properly, possible power reduction.	Short-circuit to positive.	Check wiring and connections. Replace injector if required.	1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A46 Measure point 2: Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A1 Measure point 2: Pin: 1 3- Measure type: Resistance (Ohm) Measure point 1: Pin: 1 Measure point 2: Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	Only three cylinders running.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
46	04	52	INJECTOR - INJECTOR 3 (DTC 68)	NO SIG-	Engine not working properly, possible power reduction.	Injector wiring short-circuit.	Check wiring and connections. Replace injector if required.	1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A1 Measure point 2: Pin: 1 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A31 Measure point 2: Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: Pin: 1 Measure point 2: Pin: 2 4- Measure type: Resistance (Ohm) Measure point 1: Pin: 1 Measure point 2: Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Connected; Key +15 OFF; 4- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm; 4- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	Only two cylinders running.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
46	08	52	INJECTOR - INJECTOR 3 (DTC 68)	SIGNAL NOT PLAUSIBLE	Engine not working properly, possible power reduction.	Injector not working properly.	Check wiring and connections. Replace injector if required.	1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A1 Measure point 2: Pin: 1 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A31 Measure point 2: Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: Pin: 1 Measure point 2: Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	Only three cylinders running.
47	04	52	INJECTOR - INJECTOR 3 (DTC 69)	NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring open circuit.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
47	04	52	INJECTOR - INJECTOR 3 (DTC 69)	NO SIGNAL							

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
48	01	53	INJECTOR - INJECTOR 4 (DTC 70)	EXCEEDING NORMAL RANGE	Engine not working properly, possible power reduction.	Short-circuit to positive.	Check wiring and connections. Replace injector if required.	1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A46 Measure point 2: Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A1 Measure point 2: Pin: 1 3- Measure type: Resistance (Ohm) Measure point 1: Pin: 1 Measure point 2: Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	Only three cylinders running.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
48	04	58	INJECTOR - INJECTOR 4 (DTC 70)	NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring short-circuit.	Check wiring and connections. Replace injector if required.	1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A1 Measure point 2: Pin: 1 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A31 Measure point 2: Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: Pin: 1 Measure point 2: Pin: 2 4- Measure type: Resistance (Ohm) Measure point 1: Pin: 1 Measure point 2: Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF; 4- Connector Connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm; 4- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	Only two cylinders running.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
48	08	53	INJECTOR - INJECTOR 4 (DTC 70)	SIGNAL NOT PLAUSIBLE	Engine not working properly, possible power reduction.	Injector not working properly.	Check wiring and connections. Replace injector if required.	1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A1 Measure point 2: Pin: 1 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A31 Measure point 2: Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: Pin: 1 Measure point 2: Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	Only three cylinders running.
49	04	53	INJECTOR - INJECTOR 4 (DTC 71)	NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring open circuit.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
49	04	53	INJECTOR - INJECTOR 4 (DTC 71)	NO SIGNAL							
4E	08	13	VEHICLE - CRUISE CONTROL SWITCH UNIT	SIGNAL NOT PLAUSIBLE	Cruise control / PTO not working.	Press SET+ / SET- and RESUME/ OFF at the same time.	Check correct operation of the switch by reading state parameters.				Replace wiring and connections if state does not change when Cruise Control buttons are pressed.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
50	01	94	ELECTRONIC CONTROL UNIT - MAIN RELAY DEFECT	EXCEED-ING NORMAL RANGE	Engine does not start, ECU not powered or ECU always powered and EDC off also at key-on.	Main relay interrupted or short-circuit.	Check wiring and connections. Replace relay if required.				
50	02	94	ELECTRONIC CONTROL UNIT - MAIN RELAY DEFECT	EXCEED-ING NORMAL RANGE	Engine does not start, ECU not powered or ECU always powered and EDC off also at key-on.	Main relay interrupted or short-circuit.	Check wiring and connections. Replace relay if required.				
51	01	12	VEHICLE MULTIPOSITION SELECTOR / PTO	EXCEED-ING NORMAL RANGE	Incorrect PTO operation.	Voltage exceeding max. threshold, short-circuit to positive.	Check wiring and connections. Replace sensor if required.				
51	02	12	VEHICLE MULTIPOSITION SELECTOR / PTO	EXCEED-ING NORMAL RANGE	Incorrect PTO operation.	Voltage under min. threshold, short-circuit to ground.	Check wiring and connections. Replace sensor if required.				
51	08	12	VEHICLE MULTIPOSITION SELECTOR / PTO	SIGNAL NOT PLAUSIBLE	Incorrect PTO operation.	Faulty device.	Check wiring and connections. Replace sensor if required.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
52	04	83	FUEL PRESSURE - PRESSURE MPROP REGULATOR ERROR	NO SIGNAL	Engine off.	Faulty MPROP.	Check wiring and connections. Replace MPROP if required.	Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A49 Measure point 2: ECU Pin: A19	Connector Not connected; Key +15 OFF;	Min. value: 3.2 Ohm; Max. value: 3.6 Ohm; Typical Value: 3.4 Ohm;	High noise.
52	08	83	FUEL PRESSURE - PRESSURE MPROP REGULATOR ERROR	SIGNAL NOT PLAUSIBLE			Check wiring and connections. Replace MPROP if required.	Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A49 Measure point 2: ECU Pin: A19	Connector Not connected; Key +15 OFF;	Min. value: 3.2 Ohm; Max. value: 3.6 Ohm; Typical Value: 3.4 Ohm;	
53	01	83	FUEL PRESSURE - PRESSURE MPROP REGULATOR ERROR (SHORT CIRCUIT TO POSITIVE)	EXCEEDING NORMAL RANGE		Short-circuit to battery, faulty MPROP.	Check wiring and connections. Replace MPROP if required.				
54	01	83	FUEL PRESSURE - PRESSURE MPROP REGULATOR ERROR (SHORT CIRCUIT TO NEGATIVE)	EXCEEDING NORMAL RANGE		Short-circuit to ground. Faulty MPROP.	Check wiring and connections. Replace MPROP if required.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
56	08	91	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 86)	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
5A	01	64	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 90)	EXCEEDING NORMAL RANGE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
5A	02	64	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 90)	EXCEEDING NORMAL RANGE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
5B	01	64	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 91)	EXCEED-ING NORMAL RANGE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
5E	01	27	ENGINE I - FUEL PUMP RELAY	EXCEED-ING NORMAL RANGE	Fuel pump on all-ways when engine is off.	Faulty relay, short-circuit to positive in wiring.	Turn key-on: pump must run for approximately 10 seconds (it should hum). Check pump relay if pump remains on. Check wiring if all checks are OK.				
5E	02	27	ENGINE I - FUEL PUMP RELAY	EXCEED-ING NORMAL RANGE	Fuel pump not working.	Faulty relay, short-circuit to ground in wiring.	Turn key-on: pump must run for approximately 10 seconds (it should hum). Check the pump relay, protection fuse and wiring if this does not occur.				
5E	04	27	ENGINE I - FUEL PUMP RELAY	NO SIGNAL	Fuel pump not working.	Faulty relay, wiring interrupted.	Check wiring and connections. Replace relay if required.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
5E	08	27	ENGINE I - FUEL PUMP RELAY	SIGNAL NOT PLAUSIBLE	Fuel pump not working.	Faulty relay, wiring interrupted.	Check wiring and connections. Replace relay if required.				
5F	01	82	FUEL PRESSURE - RAIL PRESSURE SENSOR OR SIGNAL ERROR	EXCEEDING NORMAL RANGE	Possible engine disconnection.	Short-circuit to positive. Faulty sensor. Rail pressure not regular.	Check wiring and connections. Replace sensor if required.				Check DTC 103 error.
5F	02	82	FUEL PRESSURE - RAIL PRESSURE SENSOR OR SIGNAL ERROR	EXCEEDING NORMAL RANGE	Possible engine disconnection.	Short-circuit to ground, faulty sensor.	Check wiring and connections. Replace sensor if required.				
60	01	82	FUEL PRESSURE - RAIL PRESSURE SENSOR OFFSET	EXCEEDING NORMAL RANGE	Possible engine disconnection.	Faulty rail pressure sensor.	Replace sensor.				
60	02	82	FUEL PRESSURE - RAIL PRESSURE SENSOR OFFSET	EXCEEDING NORMAL RANGE	Possible engine disconnection.	Faulty rail pressure sensor.	Replace sensor.				
62	01	81	FUEL PRESSURE - FAULT ON THE FUEL DRUCK CONTROL OF THE RAIL (POSITIVE DEVIATION)	EXCEEDING NORMAL RANGE	Possible engine disconnection.	High pressure circuit fuel leakage.	Check fuel feed system.				Fuel management and pressure failure in rail.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
62	01	81	FUEL PRESSURE - FAULT ON THE FUEL DRUCK CONTROL OF THE RAIL (POSITIVE DEVIATION)	EXCEED-ING NOR-MAL RANGE	Possible engine disconnection.	Injector jammed in fuel passage open position.	Check hydraulic and mechanical efficiency of injectors.				Fuel management and pressure failure in rail.
62	01	81	FUEL PRESSURE - FAULT ON THE FUEL DRUCK CONTROL OF THE RAIL (POSITIVE DEVIATION)	EXCEED-ING NOR-MAL RANGE	Possible engine disconnection.	MPROP adjuster open movement jammed.	Check efficiency of MPROP adjuster.				Fuel management and pressure failure in rail.
62	01	81	FUEL PRESSURE - FAULT ON THE FUEL DRUCK CONTROL OF THE RAIL (POSITIVE DEVIATION)	EXCEED-ING NOR-MAL RANGE	Possible engine disconnection.	Faulty high pressure pump.	Check efficiency of high pressure pump.				Fuel management and pressure failure in rail.
63	01	81	FUEL PRESSURE - FAULT ON THE FUEL DRUCK CONTROL OF THE RAIL (NEGATIVE DEVIATION)	EXCEED-ING NOR-MAL RANGE	Possible engine disconnection.	MPROP adjuster open movement jammed.	Check efficiency of MPROP adjuster.				Fuel management and pressure failure in rail.
64	01	81	FUEL PRESSURE - RAIL PRESSURE ERROR: TOO LOW	EXCEED-ING NOR-MAL RANGE	Possible engine disconnection.	High pressure circuit fuel leakage.	Check high pressure system. Replace high pressure pump if required.				Fuel management and pressure failure in rail.
65	01	81	FUEL PRESSURE - RAIL PRESSURE ERROR: TOO HIGH	EXCEED-ING NOR-MAL RANGE	Possible engine disconnection.	MPROP regulator jammed.	Check MPROP regulator, replace if required.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
66	01	81	FUEL PRESSURE - ERROR ON THE RAIL PRESSURE (EXCESSIVE DUTY CYCLE)	EXCEEDING NORMAL RANGE	Negative vehicle reaction with smoke in exhaust during acceleration.	High pressure circuit fuel leakage.	Check fuel feed system, replace high pressure pump if required. Faulty fuel feed system (fuel pump and filter jammed).				
67	01	81	FUEL PRESSURE - ERROR ON THE RAIL PRESSURE (EXCESSIVE)	EXCEEDING NORMAL RANGE	Possible engine disconnection.	MPROP regulator jammed.	Check MPROP regulator, replace if required.				Replace pressure relief valve.
68	04	96	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC I04)	NO SIGNAL		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
68	08	96	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC I04)	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
69	01	97	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY (DTC 105)	EXCEED-ING NORMAL RANGE	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.
69	02	97	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY (DTC 105)	EXCEED-ING NORMAL RANGE	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.
6A	01	97	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY (DTC 106)	EXCEED-ING NORMAL RANGE	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
6A	02	97	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY (DTC I06)	EXCEED-ING NORMAL RANGE	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.
6B	01	97	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY (DTC I07)	EXCEED-ING NORMAL RANGE	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.
6B	02	97	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY (DTC I07)	EXCEED-ING NORMAL RANGE	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
6C	01	18	VEHICLE - EDC LAMP	EXCEED-ING NOR-MAL RANGE	Warning light not working.	Short-circuit to positive.	Check correct operation of warning light using "Active diagnostic" procedure.				Warning light should come on for approximately 5 seconds at key-on. Check wiring and connections if this does not occur.
6C	02	18	VEHICLE - EDC LAMP	EXCEED-ING NOR-MAL RANGE	Warning light not working.	Short-circuit to ground.	Check correct operation of warning light using "Active diagnostic" procedure.				Warning light should come on for approximately 5 seconds at key-on. Check wiring and connections if this does not occur.
6C	04	18	VEHICLE - EDC LAMP	NO SIG-NAL	Warning light not working.	Open circuit, bulb disconnected.	Check correct operation of warning light using "Active diagnostic" procedure.				Warning light should come on for approximately 5 seconds at key-on. Check wiring and connections if this does not occur.
6C	08	18	VEHICLE - EDC LAMP	SIGNAL NOT PLAUS-IBLE	Warning light not working.	Wiring prob-lems.	Check wiring and connections. Replace sensor if required.				Warning light should come on for approximately 5 seconds at key-on. Check wiring and connections if this does not occur.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
6D	08	38	ENGINE 2 - INTERNAL ECU FAULT (PLAUSIBILITY ERROR +15)	SIGNAL NOT PLAUSIBLE			Check wiring and connections.				Key 15 off during initialisation.
6E	08	99	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC 110)	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
75	01	11	VEHICLE - VEHICLE SPEED SENSOR / SIGNAL (DTC 117)	EXCEEDING NORMAL RANGE	Maximum speed threshold has been exceeded.		Check correct calibration of speedometer.				Encourage driver to use the vehicle correctly.
75	04	11	VEHICLE - VEHICLE SPEED SENSOR / SIGNAL (DTC 117)	NO SIGNAL		Interrupted wiring between vehicle speed sensor and instrument panel.	Check wiring and connections between vehicle speed sensor and instrument panel.				
75	04	11	VEHICLE - VEHICLE SPEED SENSOR / SIGNAL (DTC 117)	NO SIGNAL		Wiring interrupted between instrument panel and EDC ECU.	Check wiring and connections between instrument panel and EDC ECU.				Intervention required if instrument panel indicates vehicle speed.

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
75	04	11	VEHICLE - VEHICLE SPEED SENSOR / SIGNAL (DTC 117)	NO SIGNAL		Vehicle speed sensor disconnected or failed.	Check correct assembly and efficiency of vehicle speed sensor.				
75	08	11	VEHICLE - VEHICLE SPEED SENSOR / SIGNAL (DTC 117)	SIGNAL NOT PLAUSIBLE		Vehicle speed sensor disconnected or failed.	Check correct assembly and efficiency of vehicle speed sensor.				
75	08	11	VEHICLE - VEHICLE SPEED SENSOR / SIGNAL (DTC 117)	SIGNAL NOT PLAUSIBLE	Vehicle speed on instrument panel does not increase sensibly.	Wrong speedometer setting.	Check correct calibration of speedometer.				
77	01	11	VEHICLE - VEHICLE SPEED SENSOR / SIGNAL (DTC 119)	EXCEEDING NORMAL RANGE	Wrong vehicle speed indication.	Wrong speedometer setting.	Check correct calibration of speedometer.				
77	02	11	VEHICLE - VEHICLE SPEED SENSOR / SIGNAL (DTC 119)	EXCEEDING NORMAL RANGE	Wrong vehicle speed indication.	Wrong speedometer setting.	Check correct calibration of speedometer.				
77	08	11	VEHICLE - VEHICLE SPEED SENSOR / SIGNAL (DTC 119)	SIGNAL NOT PLAUSIBLE	Wrong vehicle speed indication.	Wrong speedometer setting.	Check correct calibration of speedometer.				

DTC	FMI	BLINK CODE	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
79	08		ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT (DTC I21)	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

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MAIN OPERATIONS ON ENGINE MOUNTED ON VEHICLE



Keep to the following instructions before doing any work on the engine involving components of the fuel supply system.

- ☐ Before doing any work on the engine, perform the engine/vehicle fault diagnosis with specific IVECO diagnosis equipment and print out the results.
- ☐ Replacement of the MS6.3 or EDC 16 control unit must be authorized by the **Help Desk**.
- ☐ Following components in feed system cannot be overhauled but have to be replaced: pressure relief valve, if present, fuel pressure sensor, hydraulic accumulator, complete CPI high pressure feed pump, pressure control valve, electric injectors.
- ☐ All the parts of the Common Rail system are packaged by the supplier in sheets of oiled paper and are stored in cardboard boxes. They must therefore be protected against moisture and unpacked just prior to assembly.
- ☐ The greatest care must be taken over the cleanliness of parts, making sure that when handling or assembling (starting with straightforward filter and pre-filter replacement) no dirt or foreign bodies can get inside. For this reason, the plugs protecting the hydraulic parts and sensors must be removed just prior to positioning in their seats.
- ☐ Take care over the direction of assembly for all electrical connections.
- ☐ All threaded connections must be tightened to the prescribed torque.
- ☐ All the quick-coupling connectors (on the engine they are found on the high-pressure pump and on the diesel drain manifold) must be fully inserted. To drive them out, press on the tabs at the base of the connectors.

Electro-injector

None of the couplings/unions/nuts on the injector body may be handled. It is neither necessary nor permitted to dismantle the nozzle body or the electromagnet.

If working on the high-pressure pipe, the hexagon on the injector side must be kept stationary with a wrench.

Before working on pipes, make sure the injector is stationary in its seat on the cylinder head.

When assembling/disassembling the injector drain, the retaining spring must not be removed from its seat in the injector: pushing the spring towards the engine and applying a vertical force on the connector frees the recirculation. When assembling, rest the recirculation connector in its seat and apply a vertical force while keeping the retaining spring pressed in the direction of the engine. Fitting in has to be easy.

CP3 High-pressure pump

If working on the high-pressure pipe, the hexagon on the pump side must be kept stationary with a wrench.

Before working on the high-pressure pipe, make sure the pump is secured in its seat.

High-pressure pipes

Each high-pressure pipe must be replaced after disassembly operations.

The couplings must be tightened or loosened with the injectors, hydraulic accumulator (rail) and high-pressure pump well secured and taking care to keep the hexagon on the component side stationary, space permitting.

Hydraulic accumulator (rail) and accessories

The pressure sensor can be assembled five consecutive times; after that, it must be replaced.

They must be lubricated with a thin layer of oil before being mounted.

540110 ENGINE REMOVAL-REFITTING

Removal

Place the vehicle in the pit or on an auto lift.

Lift engine hood (2), remove fastening screws (1), then remove hood (2).

Remove the supporting rod.

Disconnect negative cable (7) from battery (6).

Disengage cable (11) from the hood opening device.

Disconnect front headlamp electric connections (10 and 13).

Remove nuts (8), screws (9), then remove front cross-member (11) complete with the headlamps.

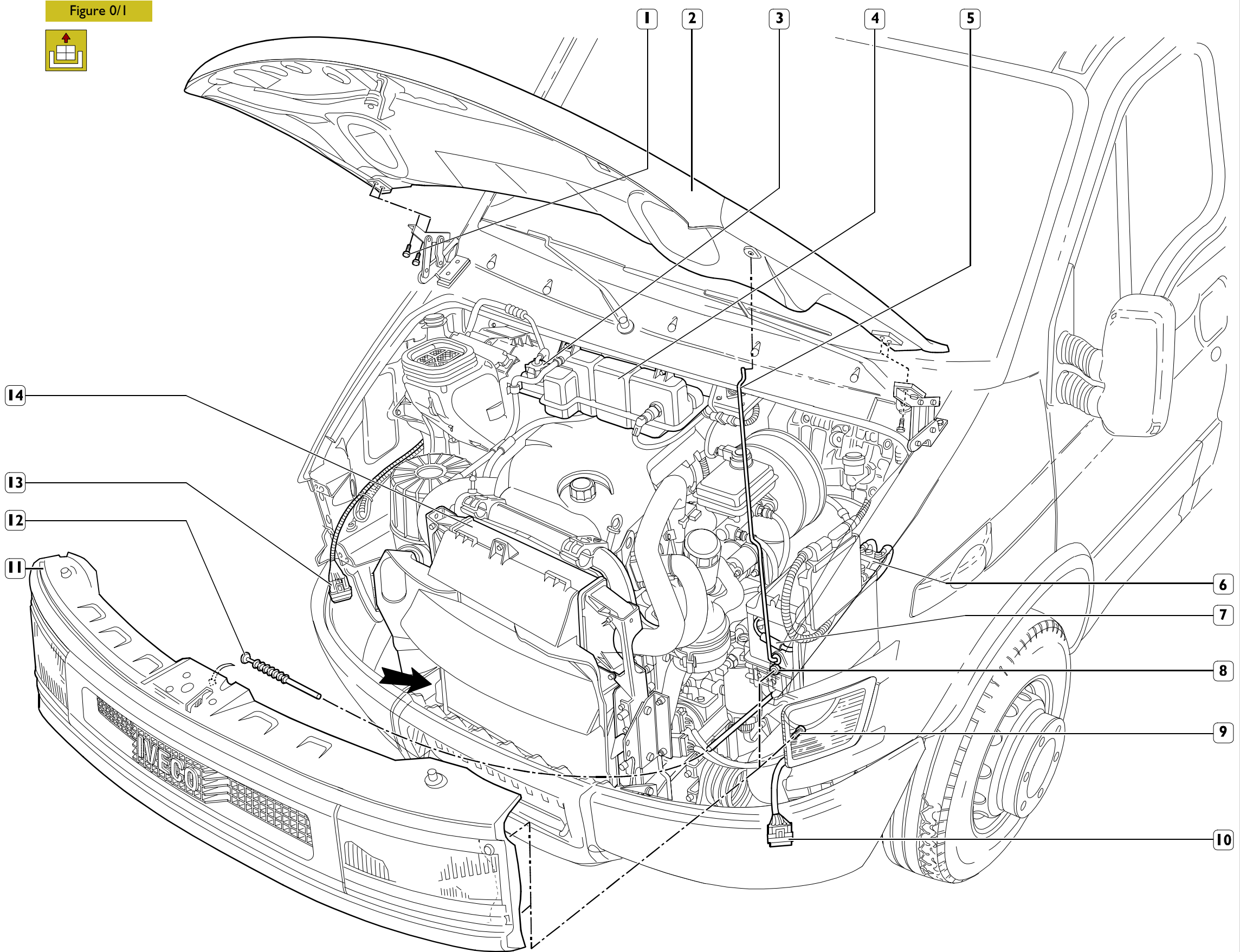
Remove cap (3) from expansion tank (4).

Carry out the following operations, from under the vehicle (see Figure 0/4):

- ☐ remove screws (15), and then take off central shelf (12);
- ☐ remove screws (8 and 10), then take off side guards (11).

Place a suitable container under radiator (14), remove plug (→), located on the right side of the radiator, then drain the engine coolant.

Figure 0/1



Loosen the straps, then remove air pipe (6) from heat exchanger (11) and (12), from the inlet manifold.

Loosen the straps, then disconnect the following coolant vent ducts: (22), from the cylinder head; (24), from radiator inlet pipe (8); (23), from radiator (10).

Loosen the straps, then disconnect the following coolant pipes: (13), from the heat exchanger; (28), from the thermostat body.

Remove screws (7) fastening air baffle (9) to radiator (10).

Remove screws (15) fastening radiator (10) to the supports. Disengage, from the left side, electric connection (16) supporting bracket (14).

Remove, by properly turning air baffle (9), radiator (10) complete with the heat exchanger, from the supports.

NOTE In case of vehicles equipped with cabin air-conditioner, proceed as follows:

- ☐ blow gas off the air-conditioning system, as described in the relevant chapter in the "Bodywork and chassis" section;
- ☐ disconnect the electric connection from the drying filter;
- ☐ disconnect the pipes from the condenser, and then seal the pipes and their respective fittings, to prevent moisture and impurities from penetrating into the system.

From water pump pipe (26), disconnect both expansion tank (17) pipe (25) and cabin heater pipe (1).

Disconnect the following electric connections: (19), from the pressure switch; (27), from level sensor (27) located on expansion tank (17).

Remove expansion tank (17) fastening nuts (18), and then take the expansion tank off the wall.

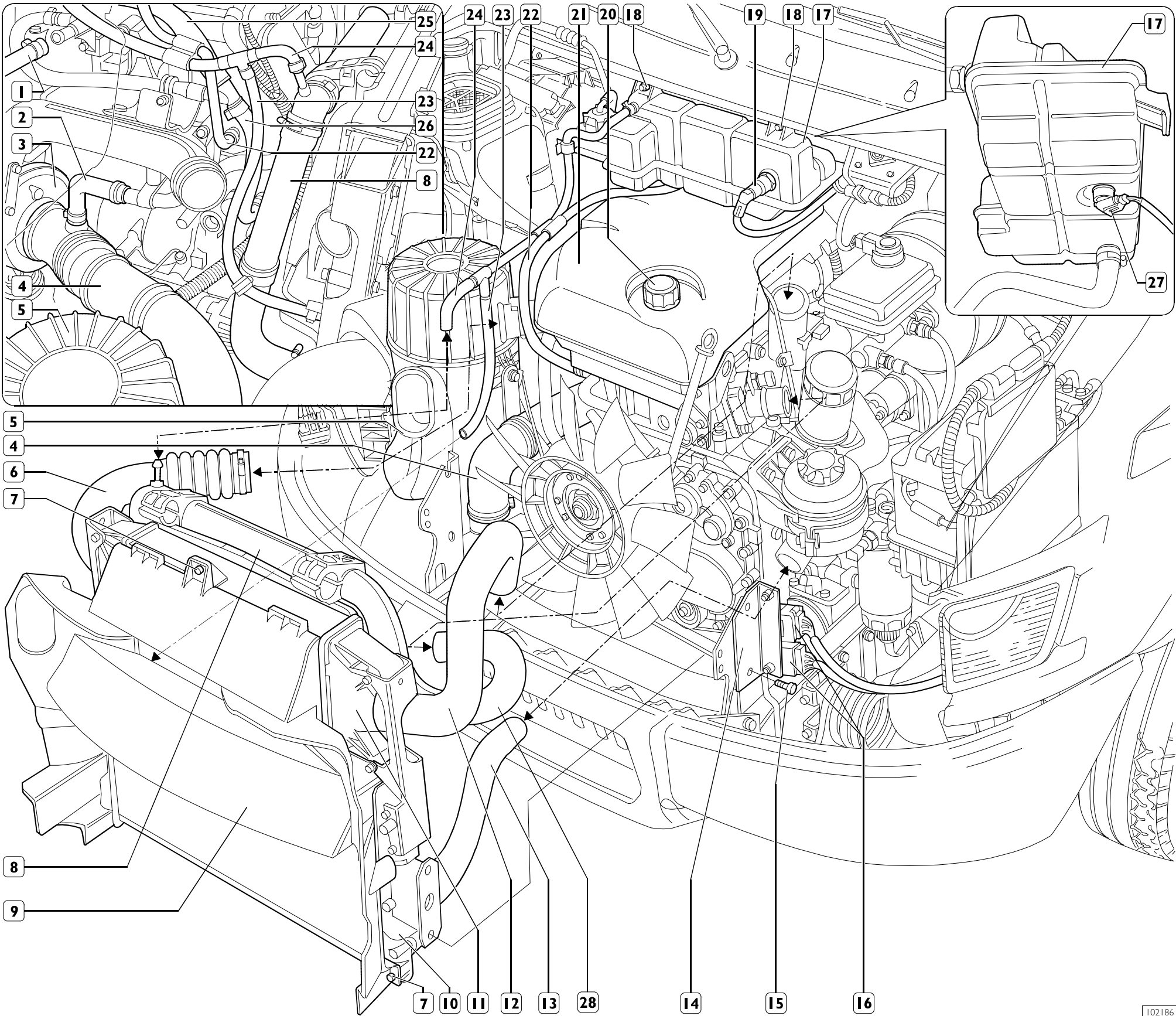
Remove oil filling plug (20), then take off sound-proofing cover (21).

Loosen the strap, then disconnect pipe (2) from its respective oil vapour recirculation pipe.

Loosen the straps, then remove air duct (4) both from turbocharger (3) and air filter (5).

NOTE As regards the engines equipped with variable-geometry turbochargers, properly plug the turbocharger air outlet, in order to prevent foreign bodies from getting into accidentally.

Figure 0/2



Disconnect vacuum pipe (3) from variable-geometry turbocharger actuator (2) (where available).

Cut the straps securing the wiring to the engine, then disconnect the following electric connections: VGT solenoid valve (6, where available), alternator (4), oil level sensor (7), engine revs sensor (8), electro-magnetic joint (9), coolant temperature sensors (22), phase sensor (10), electric injectors (26), air temperature sensor (23), pre-heating spark plugs (24), fuel pressure sensor (25), fuel pressure regulator (20), oil pressure sensor (16).

Remove the screw, and then take off earth cable (14).

Remove the nuts, and then remove the cables from starting motor (17).

Disconnect power brake vacuum pipe (15) from union (11), from the low-pressure fuel pipes, then disconnect both supply pipe (18) and return pipe (19).

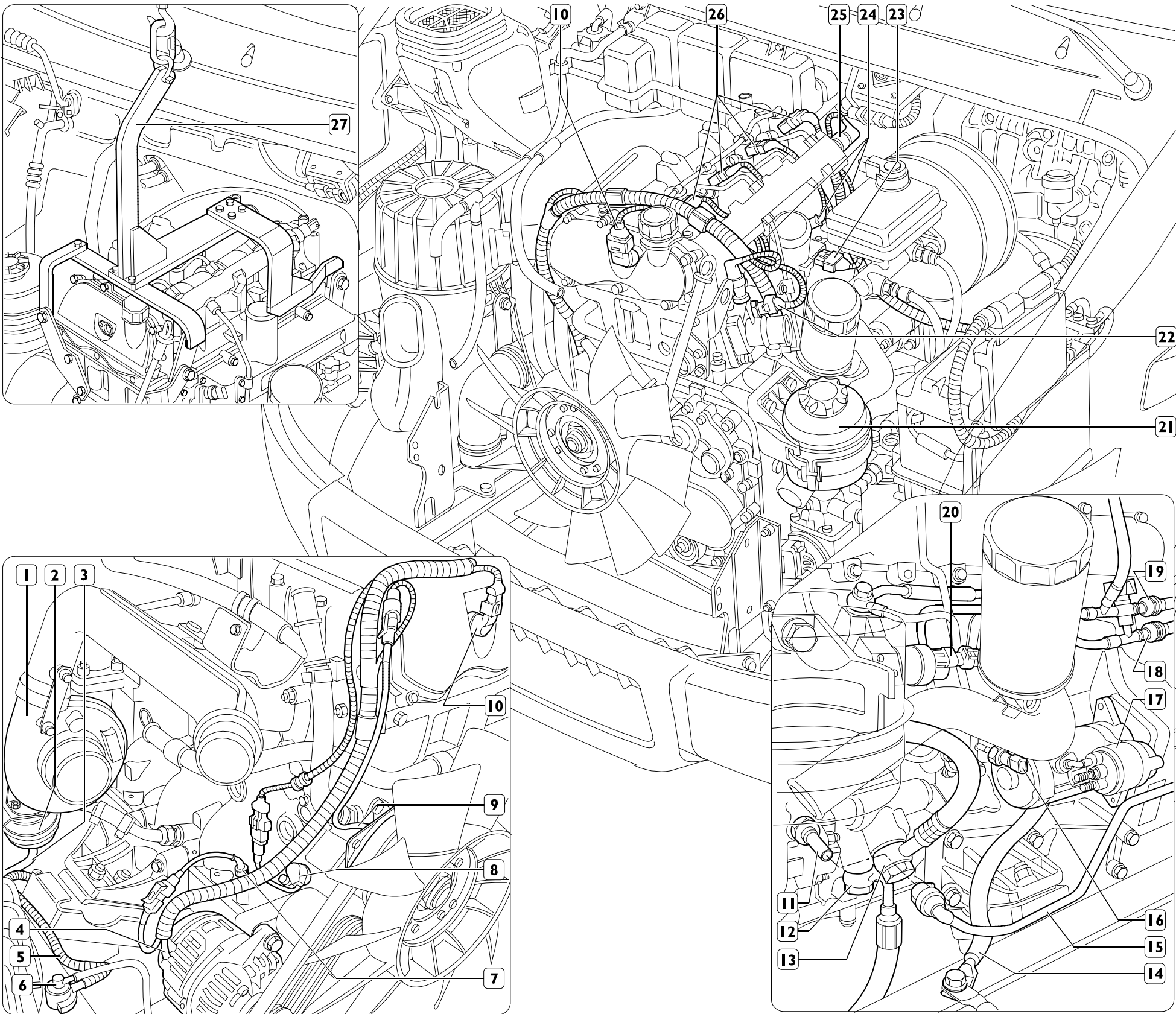
Place a container under the power steering pump to recover the oil from the system, and then disconnect oil supply and delivery pipes (12 and 13).

Loosen the fastening strap, then remove power steering tank (2) from the support.

Apply tool 99360543 (27) to the brackets of the engine to remove the latter from its compartment, and then secure it to the hoist.

NOTE In case of vehicles equipped with cabin air-conditioning system, disconnect the cooling gas pipes from the compressor, then seal the pipes and their respective fittings on the compressor, to prevent moisture and impurities from penetrating into the system.

Figure 0/3



- ❑ Disconnect the screws (16 and 20) securing the brackets (17 and 19) and disconnect the "bowdens" (18 and 21) from the gearbox.
- ❑ Unscrew the fixing screws (22), move the clutch control cylinder (23), with its bracket, and fasten it to the chassis frame appropriately.
- ❑ Remove the sealing from the ring nut (1), unscrew it and disconnect the speedometer control cable.
- ❑ Disconnect the electrical connection (4) from the reversing light switch.
- ❑ Disconnect the exhaust pipe (9) from the turbocharger outlet pipe.
- ❑ Put a jack under the gearbox to support it.
- ❑ Disconnect the bracket supporting the gearbox on the rear crosspiece by undoing the four screws (5).
- ❑ Unscrew the fixing screws (6) and remove the crosspiece (7) supporting the gearbox complete with the gearbox/support bracket.
- ❑ Remove nuts (14) securing elastic supports (13) to the chassis.
- ❑ Remove bolts (3) securing drive shaft (2) to the gear shift. Remove, if necessary, screws (24) securing elastic support (25) to the chassis, then properly secure the drive shaft to the chassis.
- ❑ Take the jack out from under the gearbox.
- ❑ Lift the engine assembly and take it out of the engine bay.

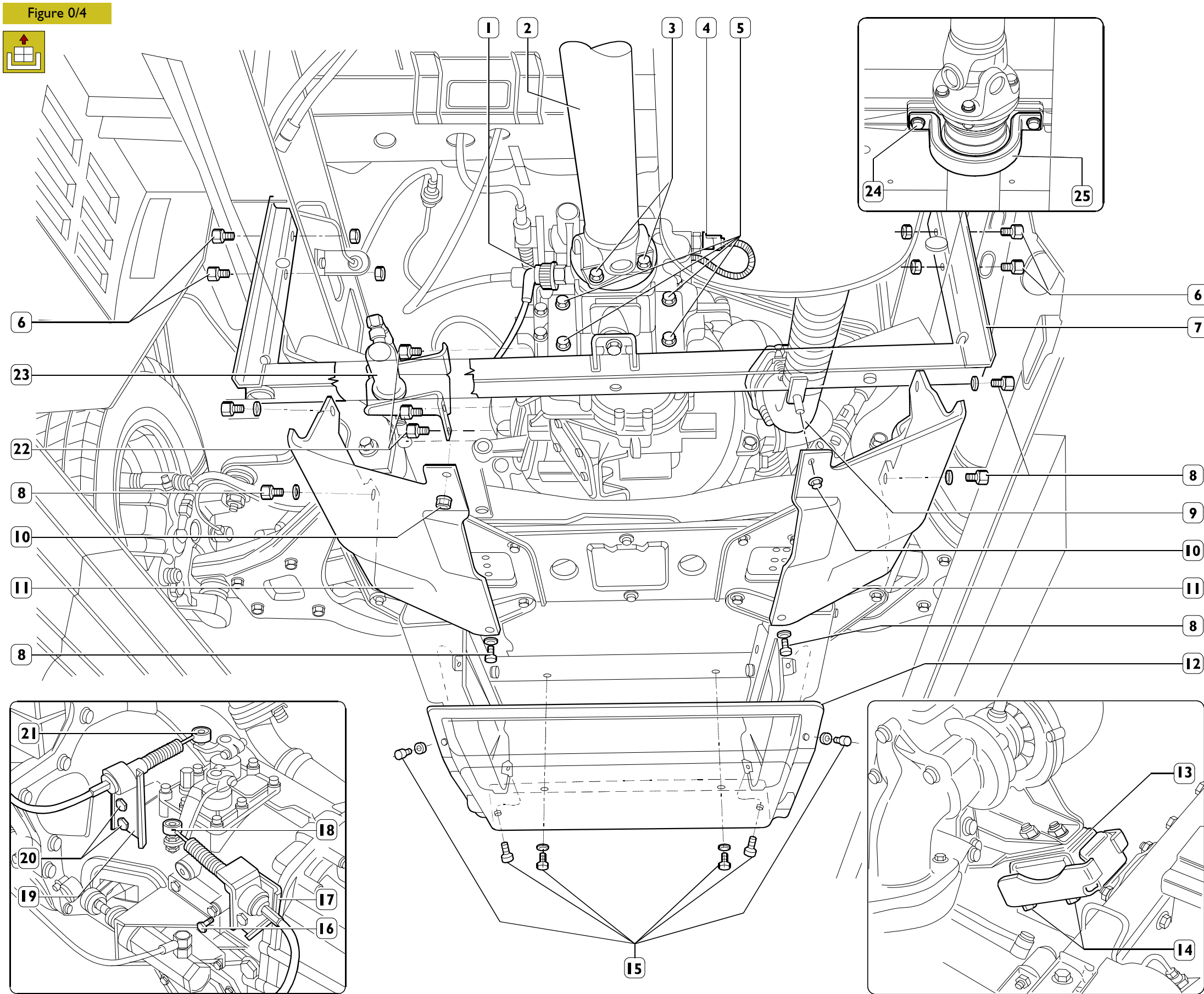
NOTE The power unit must be removed from the engine compartment with the greatest care, to avoid damaging the remaining parts on the vehicle, in particular the steering box oil pipes.

If it is necessary to detach the gearbox from the engine, take out the fixing screws and remove the starter motor.

Take out the fixing screws and detach the gearbox from the engine.

Should this prove difficult, take the inspection cover off the front cover of the gearbox.

Using special pliers, open out the circlip retaining the thrust-bearing sleeve to the clutch plate while detaching the gearbox from the engine.





Refitting

To refit the engine assembly, carry out the operations described for removal in reverse order, following these instructions:

- ☐ Before refitting the gearbox to the engine, it is necessary to remove the pressure plate bearing from the diaphragm spring by opening out the retaining circlip. Fit the pressure plate bearing on the sleeve of the drive input shaft cover, connecting it to the clutch release lever. Spread the gearbox input shaft with Molikote molybdenum disulphide grease. Engage a gear to let the main shaft turn, rotating the propeller shaft connecting flange. Push the gearbox fully in so that the pressure plate bearing couples with the diaphragm spring correctly.
- ☐ Pay special attention to the operations needed to install the engine assembly in the engine bay.
- ☐ Check the conditions of the coolant pipes or sleeves and of the air ducts. Replace them if they show any sign of deterioration.
- ☐ Check the flexible mountings of the assemblies: engine and gearbox. Replace them if they show any sign of deterioration.
- ☐ Check that the exhaust pipe members have not deteriorated and are not about to deteriorate. If this is so, replace them along with the flexible parts for securing them.
- ☐ Tighten the screws or nuts to the required torque.
- ☐ Meticulously check the state of the vacuum pipe. It must show no sign of cracking, cutting, scoring or of being crushed. Replace it if there is any doubt at all about its soundness. When mounting it, make sure the pipe does not come into contact with sharp metal parts or corners or with any particularly hot parts. In addition, after assembly, the pipe must have no bends or constrictions, its radius of curvature should be broad and it must be secured to the vacuum pump fitting with a suitable clamp.
- ☐ Make sure the quick-coupling fittings of the fuel pipes are thoroughly clean and, after connection to the relevant high-pressure pump unions or fuel filter mount, are fully inserted and do not come loose.
- ☐ Fill the cooling system with coolant.
- ☐ Fill the hydraulic power steering circuit and bleed the air as described under the relevant heading.
- ☐ Check the level of oil in the engine and gearbox.
- ☐ Recharge the air-conditioning system (if any), as described in the relevant chapter in the "Bodywork and chassis" section.

NOTE When positioning the engine in the engine bay, take special care not to damage the top pipe of the power steering and the soundproof-heatproof cladding of the engine bay. Once positioned, meticulously check that the top pipe of the power steering is sound. Before using it again, check that the power steering oil and coolant contain no impurities. If they do, filter with suitable mesh filters. For any topping up, refer to the REPLENISHING FLUIDS table in the "GENERAL" section.

Checks and tests



Start up the engine, leave it running just a little faster than idling speed and wait for the coolant temperature to reach the value for opening the thermostat, then check that:



- ☐ No water leaks from the connecting sleeves of the engine cooling and cab heating circuit pipes; tighten the collars if necessary.
- ☐ No oil leaks from between the cover and cylinder head, oil sump and crankcase, oil filter and its seat, heat exchanger and crankcase or from between the various pipes of the lubricating circuit.
- ☐ No fuel leaks from injection pump and injector lines. Tighten fittings if necessary.
- ☐ Check the indicator and warning lights on the instrument panel and the devices disconnected on removing the engine all work properly.



501430 Power steering system air bleed

Check the level of oil in the tank and top it up if necessary. Lift the vehicle at the front, start up the engine and let it idle for some time.

Check there is no oil leakage from the hydraulic circuit and check the level in the tank.

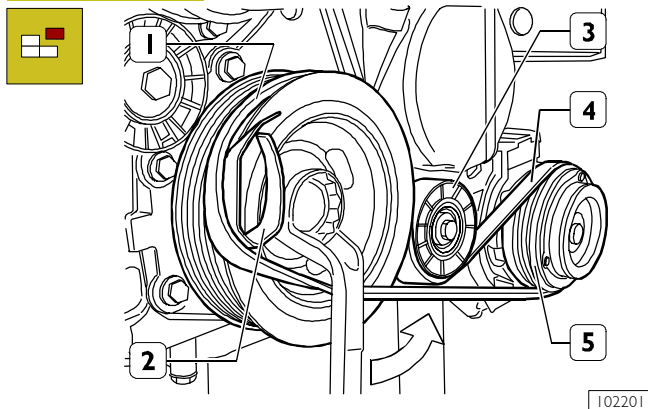
Slowly turn the steering wheel in both directions of steering so that the air in the hydraulic system comes out.

Check the level of oil in the tank again and top up if necessary.

543910 REPLACING AIR-CONDITIONING COMPRESSOR DRIVE BELT (VERSION WITH BELT TENSIONER)

Disassembly

Figure 0/5



Place the vehicle in a pit or on an auto lift. Remove, from under the vehicle, the central sound-proofing guard. Remove elastic belt (4) from pulleys (1 and 5).

Assembly

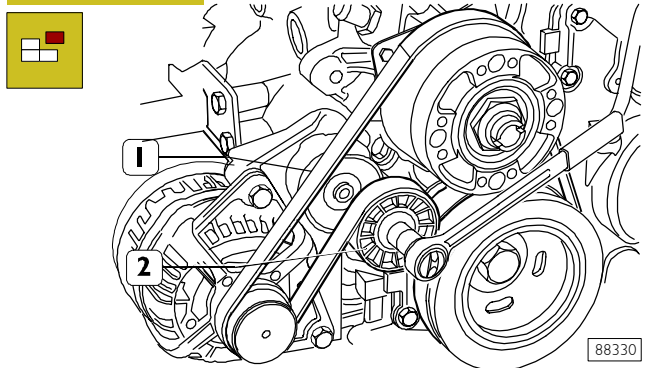
Apply wedge 99360161 (2) with elastic belt (4) to pulley (1), then place the elastic belt onto roller (3) and pulley (5), taking care to place the pulley ribs in the corresponding races of pulleys (1 and 5).

Rotate the drive shaft counterclockwise (→) until belt (4) is correctly coupled with pulley (1).

543910 POWER STEERING PUMP-ALTERNATOR BELT REPLACEMENT

Disassembly

Figure 0/6



Disassemble the compressor drive belt, if there is one, as described under the relevant heading. Slacken off the tension of the belt (1) using a specific wrench on the automatic tightener (2) and remove the belt.

Assembly

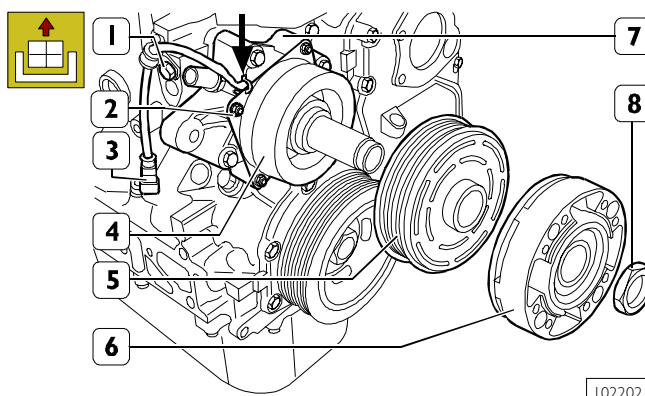
Mount the drive belt (1) taking care to position its ribs correctly in the respective races of the pulleys. Release the automatic tightener (2). Turn the crankshaft by one turn to settle the belt.

Mount the compressor drive belt, if there is one, and adjust the tension as described under the relevant heading. Fit the middle soundproofing guard back on.

543210 REPLACING THE WATER PUMP

Removal

Figure 0/7



Drain the cooling fluid, and then remove both the front cross-member and the radiator, as described in the "Power unit detachment/re-attachment" chapter. Take the fan off the electro-magnetic joint. Remove electric connection (3) from the engine cable. Stop rotation of electro-magnetic joint (6), then remove nut (8).

NOTE Unscrew nut (8) in a clockwise direction, since the nut thread goes leftwards.

Take off hub (6) and pulley (5).

Cut the strap (→), remove electric cable (3) retaining strap fastening screw (1), remove nuts (2), then take electro-magnet (4) off water pump (7). Remove the fastening screws, and then take off water pump (7).

Refitting



Re-attachment is carried out by reversing the order of detachment operations. In particular, tighten the screws and nuts to the specified torque values.



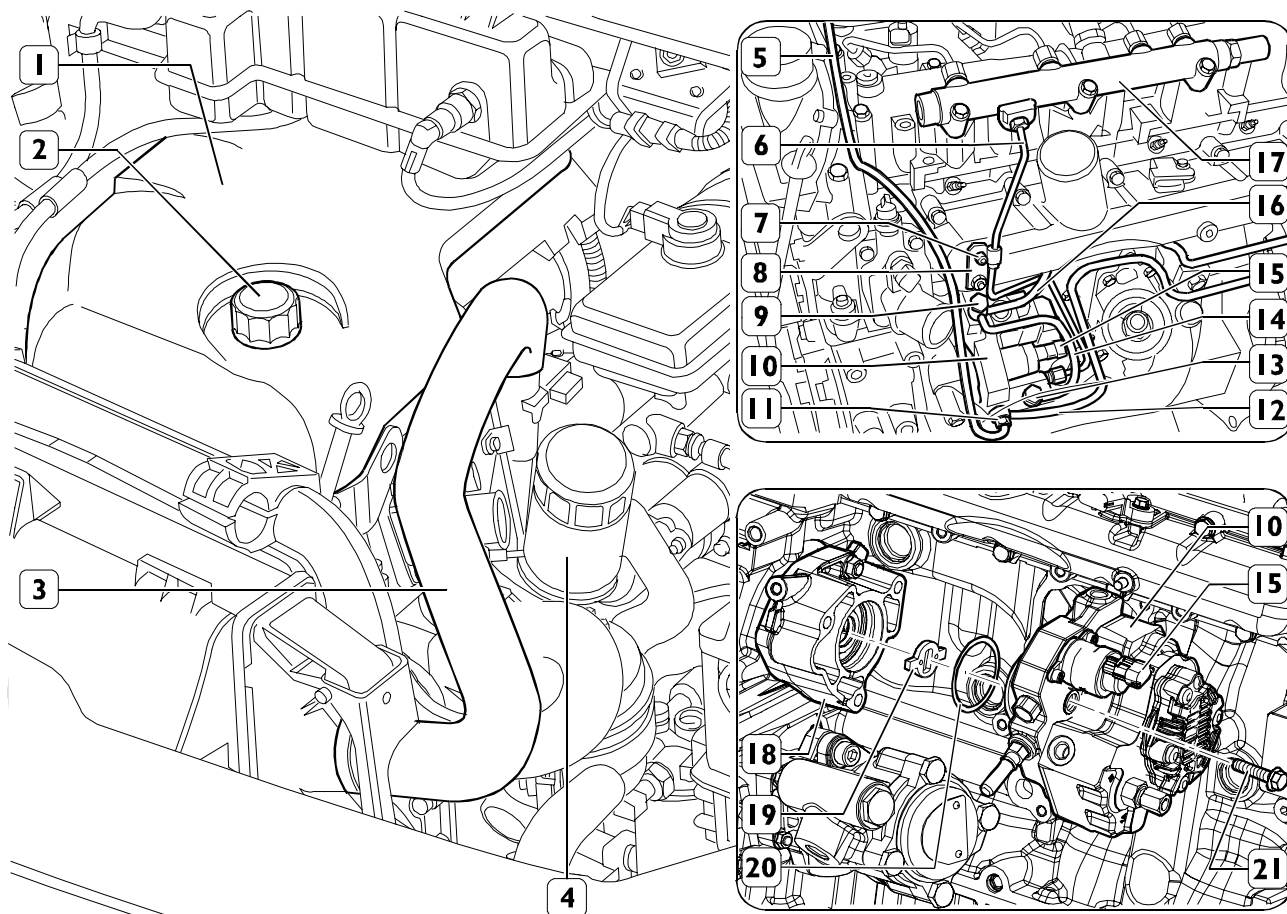
After re-attachment has been completed, fill the engine cooling system, start the engine and check for cooling fluid leaks.



771010 REPLACING THE HIGH-PRESSURE PUMP

Removal

Figure 0/8



102203

Remove plug (2), then take cover (1) off the cylinder head. Loosen the straps, then remove air duct (3) both from the inlet manifold and the heat exchanger.

Use tool 99360076 to take oil filter (4) off the heat exchanger.

Remove the electric connection from pressure regulator (15).

Remove fittings (9, 11 and 13), then disconnect low-pressure pipes (5, 12, 14 and 16) from high-pressure pump (10).

Remove pipe (6) retaining bracket (8) fastening screw (7), and then disconnect the pipe both from hydraulic accumulator (17) and high-pressure pump (10).

Remove screws (21), and then take high-pressure pump (10) off support (18). Remove joint (19).

Refitting



Re-attachment is carried out by reversing the order of detachment operations. In particular, comply with the following instructions:



☐ replace the seal rings, gaskets, high-pressure pipe and oil filter with new parts;



☐ lubricate, prior to assembling, seal ring (20) with engine oil;



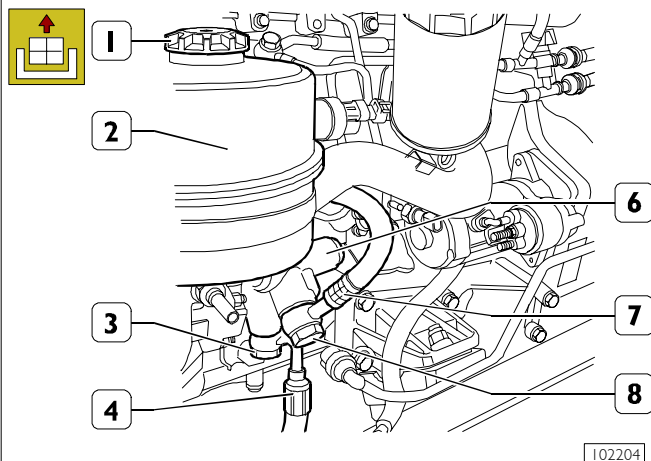
☐ tighten the screws, nuts, fittings and oil filter to the specified torque value;

☐ check the engine oil level; top up, if necessary.

501450 REPLACING THE POWER STEERING PUMP

Removal

Figure 0/9



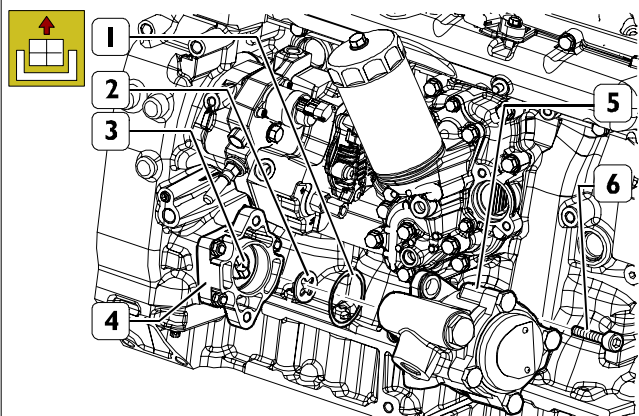
102204

Remove oil filling plug (1) from oil tank (2).

Remove, from under the engine compartment, the central sound-proofing guard.

Place a container under power steering pump (6) to recover the oil from the system, and then remove fittings (3 and 8) and disconnect oil pipes (4 and 7) from power steering pump (6).

Figure 0/10



102205

Remove screws (6) and take power steering pump (5) off support (4). Remove joint (2) from drive arbor (3).

Refitting



Re-attachment is carried out by reversing the order of detachment operations. In particular, take care of the following:



☐ replace seal ring (1) with a new part, and lubricate the same with engine oil prior to assembling;



☐ replace the oil pipe gaskets (as above);
☐ tighten the nuts, screws and fittings to the specified torque values.

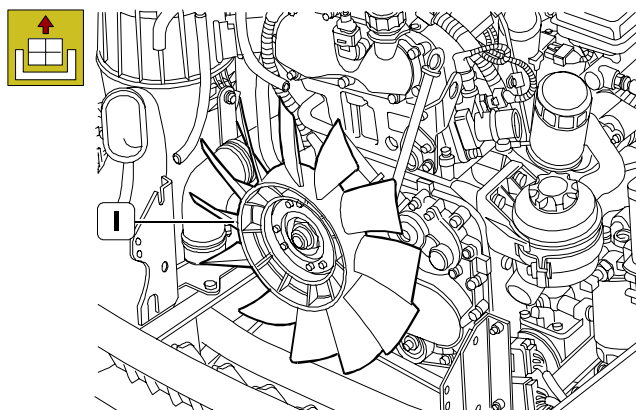


After re-attachment has been completed, fill the hydraulic power steering circuit and bleed the air as described under the relevant heading.

540440 REPLACING THE DRIVE SHAFT SEAL RING AND THE FRONT COVER GASKET

Removal

Figure 0/11



102206

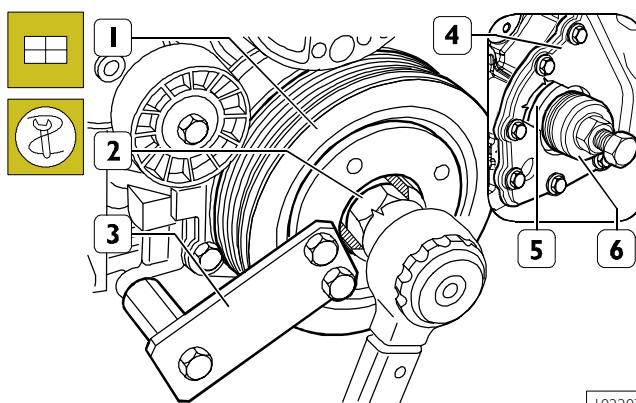
Proceed as follows, by operating as described in the "Power unit detachment/re-attachment" chapter:

- ☐ drain the engine cooling fluid;
- ☐ remove the front cross-member;
- ☐ remove the radiator.

Remove fan (1) from the electro-magnetic joint.

Take off the air-conditioning compressor drive belt (if any) as well as the water pump-alternator drive belt, as described in the relevant chapters.

Figure 0/12



102207

Use tool 99360190 (3) to stop drive shaft rotation, then remove screw (2) and take off damper pulley (1).

Apply tool 99340059 (6) as shown in the figure to remove seal ring (5) from cover (4).